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**Release 2**

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# Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Intelligent Transport Systems (ITS).

The present document is part 2 of a multi-part deliverable. Full details of the entire series can be found in part 1 [i.1].

# Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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# Executive summary

The present document is based on the use cases defined in ETSI TR 103 300-1 [i.1].

The first major clause, clause 4, provides background information and a further analysis of the use cases. It recommends information flows for VRU use cases for which a specific Vulnerable Road User Awareness Message (VAM) should be defined, and presents a security analysis of the use cases defined in ETSI TR 103 300-1 [i.1]. The security analysis is used in clause 5 to derive security requirements.

Clause 5 presents formal functional and operational requirements for system, communications, and security.

Clause 6 outlines the functional architecture of the VRU system. The following is specified:

- VRU device profiles:
  - VRU profiles classify VRUs based on their typical behaviour. The profiles defined in the present document cover pedestrians, cyclists, motorcyclists, and animals. The aim is to allow implementers to target specific VRU profiles with their devices, so that they can produce specialized devices at lower cost.
- VRU clustering:
  - VRU clustering is a technique used to reduce traffic on the airlink by allowing VRU devices, under particular conditions, to delegate responsibility for sending VAMs to a device known as the "cluster leader", which is associated with a different VRU within the cluster.
- The services that are used by a VRU application running on a VRU ITS-S and how they are to be situated within the ITS-S architecture.
- The uses of the different interfaces with a VRU system.

Finally, the present document describes any necessary changes to the ITS-S application and facilities layers, and to the Station Management and Security Management entities within the ITS-S.

# Introduction

VRU applications extend the awareness of Vulnerable Road Users and about Vulnerable Road Users to all road users. The objective is to protect road users such as motorcycles, bicycles, pedestrians and more impaired traffic participants in the neighbourhood of other traffic participants. They enable further improvement of traffic safety and management based on both direct ITS station-to-ITS station communications and via a third party ITS station (e.g. vehicle, roadside equipment or central ITS station).

The present document specifies the VRU system requirements and architecture building from the analysis of the use cases described in ETSI TR 103 300-1 [i.1]. The specification of the VRU system requirements and architecture leads to recommendations for updates of relevant C-ITS standards.

# 1 Scope

The present document analyses the impact of use cases described in ETSI TR 103 300-1 [i.1] and specifies the VRU related requirements, as well as the functional architecture of the VRU system that will prevent collisions with other road users. In addition, it provides the impact of the specified requirements and functional architecture on relevant C-ITS standards, identifying which messages are needed to support the use cases described in ETSI TR 103 300-1 [i.1].

# 2 References

## 2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI EN 302 665: "Intelligent Transport Systems (ITS); Communications Architecture".

[2] ETSI TS 102 894-1: "Intelligent Transport Systems (ITS); Users and applications requirements; Part 1: Facility layer structure, functional requirements and specifications".

[3] ETSI EN 302 637-2: "Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Part 2: Specification of Cooperative Awareness Basic Service".

[4] ETSI EN 302 637-3: "Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Part 3: Specifications of Decentralized Environmental Notification Basic Service".

[5] ETSI EN 302 890-2: "Intelligent Transport Systems (ITS); Facilities Layer function; Part 2: Position and Time management (PoTi); Release 2".

[6] ETSI TS 103 097: "Intelligent Transport Systems (ITS); Security; Security header and certificate formats".

[7] ETSI TS 101 539-1: "Intelligent Transport Systems (ITS); V2X Applications; Part 1: Road Hazard Signalling (RHS) application requirements specification".

## 2.2 Informative references

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NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ETSI TR 103 300-1: "Intelligent Transport System (ITS); Vulnerable Road Users (VRU) awareness; Part 1: Use Cases definition; Release 2".

[i.2] Deliverable D4.2: "Architecture for integration of VRUs and draft recommended practices for usability & user acceptance", Project H2020 VRUITS, December 2014.

[i.3] SAE Surface Vehicle Standard J2945/5: "Service Specific Permissions and Security Guidelines for Connected Vehicle Applications".

[i.4] ETSI TR 103 562: "Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Analysis of the Collective Perception Service (CPS)".

[i.5] C-ITS Platform, Final report Phase II, September 2017.

[i.6] SAE J2945/9™ (March 2017): "Vulnerable Road User Safety Message Minimum Performance Requirements".

[i.7] ETSI TS 102 165-1: "CYBER; Methods and protocols; Part 1: Method and pro forma for Threat, Vulnerability, Risk Analysis (TVRA)".

[i.8] IETF RFC 8446: "The Transport Layer Security (TLS) Protocol Version 1.3".

NOTE: Available from https://tools.ietf.org/html/rfc8446.

[i.9] IETF RFC 5246: "The Transport Layer Security (TLS) Protocol Version 1.2".

NOTE: Available from https://tools.ietf.org/html/rfc5246.

[i.10] ETSI TS 102 940: "Intelligent Transport Systems (ITS); Security; ITS communications security architecture and security management".

[i.11] ETSI TS 103 300-3: "Intelligent Transport Systems (ITS); Vulnerable Road Users (VRU) awareness; Part 3: Specification of VRU awareness basic service; Release 2".

[i.12] ETSI EN 302 890-1: "Intelligent Transport Systems (ITS); Facilities Layer function; Part 1: Services Announcement (SA) specification".

[i.13] ETSI TS 103 301: "Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Facilities layer protocols and communication requirements for infrastructure services; Release 2".

[i.14] ETSI TS 102 894-2 (V1.3.1): "Intelligent Transport Systems (ITS); Users and applications requirements; Part 2: Applications and facilities layer common data dictionary".

[i.15] Regulation (EU) No 168/2013 of the European Parliament and of the Council of 15 January 2013 on the approval and market surveillance of two- or three-wheel vehicles and quadricycles.

NOTE: Available at http://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32013R0168&from=EN.

[i.16] ISO/IEC 27001:2013: "Information technology -- Security techniques -- Information security management systems -- Requirements".

[i.17] FIPS PUB 199: "Standards for Security Categorization of Federal Information and Information Systems".

[i.18] ETSI TS 103 175: "Intelligent Transport Systems (ITS); Cross Layer DCC Management Entity for operation in the ITS G5A and ITS G5B medium".

[i.19] SAE J3016™: "Levels of Driving Automation".

[i.20] Yingying Zhang, Danya Yao, Tony Qiu, Lihui Peng: "Scene-based pedestrian safety performance model in mixed traffic situation", IET Intelligent Transportation Systems 2014, Vol. 8 Issue 3, pp 209-218.

[i.21] Ahmed Tageldin, Mohamed Zaki, Tarek Sayed: "Examining pedestrian evasive actions as a potential indicator for traffic conflicts", IET Intelligent Transportation Systems 2017, Vol. 11 Issue 5, pp 282-289.

[i.22] European Data Protection Board: "Guidelines 1/2020 on processing personal data in the context of connected vehicles and mobility related applications Version 1.0", January 2020.

[i.23] ETSI TS 102 731: "Intelligent Transport Systems (ITS); Security; Security Services and Architecture".

[i.24] CEN ISO/TS 17423: "Intelligent transport systems -- Cooperative systems -- ITS application requirements and objectives for selection of communication profiles".

[i.25] IEEE 802.11™-19/0495r3: "802.11bd Functional Requirements Document", March 2019.

NOTE: Available at https://mentor.ieee.org/802.11/dcn/19/11-19-0495-03-00bd-802-11bd-functional-requirements-document.doc.

[i.26] IEEE 1609.2™: "IEEE Standard for Wireless Access in Vehicular Environments--Security Services for Applications and Management Messages".

[i.27] ETSI TS 103 248: "Intelligent Transport Systems (ITS); GeoNetworking; Port Numbers for the Basic Transport Protocol (BTP); Release 2".

# 3 Definition of terms, symbols and abbreviations

## 3.1 Terms

For the purposes of the present document, the terms given in ETSI EN 302 665 [1], ETSI TS 102 731 [i.23], ETSI TR 103 300-1 [i.1] and the following apply:

**actor:** participant in a use case

**attestation:** process by which a device proves to another party that it is in a known valid state

**authorization authority:** security management entity responsible for issuing, monitoring the use of, and withdrawing authorization tickets (from ETSI TS 102 731 [i.23])

**authorization ticket:** data object that demonstrates that the valid holder is entitled to take specific actions (from ETSI TS 102 731 [i.23])

**availability:** security service providing assurance that intended recipients of certain data will in fact receive that data

**blacklisting:** process by which an authorization authority is instructed not to issue authorization tickets to an end entity in future

**broadcast:** transmission method used to send facilities layer messages to all endpoints within the specified communication range

NOTE: As defined in CEN ISO/TS 17423 [i.24], broadcast is one of the possible destination types that can be specified by an application without any reference to the type of access technology. How broadcast is achieved in lower layers is out of scope of the present document.

**certificate:** type of authorization ticket; a digital document issued by a trusted third-party, binding properties (identity or permissions) to a cryptographic public key, so that use of the corresponding private key (typically to sign a message) gives assurance that the private key holder has the indicated properties

**combined VRU:** combination of a VRU and a VRU vehicle (e.g. bicycle, wheel-chair)

**communication range:** distance over which an ITS-S can act in the indicated role based on successful transmission

NOTE: How to achieve the communication range using the lower layers is out of scope of the present document.

**confidentiality:** security service providing assurance that unauthorized parties cannot read certain data

**dynamic state:** collection of properties of a moving object including position, velocity, acceleration, mass, and forces acting on the object, that allow path prediction

**end-entity:** user of an authorization ticket

**heterogeneous VRU cluster:** VRU cluster composed of VRU belonging to different VRU profiles

**homogeneous VRU cluster:** VRU cluster composed of VRU belonging to the same VRU profile

**integrity:** security service providing, in the context of the present document, assurance that certain data and associated metadata are correct

NOTE: This term has a narrower meaning in other contexts, but the definition given is the definition used in the present document.

**Lateral Distance (LaD):** estimated distance between the VRU and the vehicle perpendicular to the direction of the vehicle heading

**Longitudinal Distance (LoD):** estimated distance between the VRU and the vehicle along the direction of the vehicle heading

**Manoeuvre Identifier (MI):** identifier of a manoeuvre used in a Manoeuvre Coordination Service (MCS)

NOTE: The choice of manoeuvre may be generated locally based on the available sensor data at the VRU ITS-S and may be shared with neighbouring ITS-S (VRUs or non-VRUs) in the vicinity to initiate a joint manoeuvre coordination among VRUs. See also clause 6.5.10.9.

**Minimum Safe Lateral Distance (MSLaD):** minimum lateral separation between the VRU and vehicle for safety

**Minimum Safe Longitudinal Distance (MSLoD):** minimum longitudinal separation between the VRU and the vehicle for safety

**Minimum Safe Vertical Distance (MSVD):** minimum vertical separation between the VRU and the vehicle for safety

**misbehaviour:** sending messages within an application that cause undesirable actions to be taken by the receivers

**misbehaviour detection:** process whereby a receiver of messages determines that those messages constitute misbehaviour

**misbehaviour reporting:** process whereby an actor that has carried out misbehaviour detection, reports the misbehaving messages to a central authority to determine whether the originating device should be blacklisted

**pseudonymity:** security service providing protection against an unauthorized party being able to determine that multiple different messages have come from the same source

NOTE: This service is only needed when confidentiality services are not applied.

**role:** property of an actor in a system indicating what activities they are entitled to carry out or request within that system

**security control:** feature of a system included in order to achieve a security goal of the system

**security goal:** system requirement to ensure that the system has the appropriate security properties

**security property:** one of confidentiality, integrity, availability and pseudonymity

**sensitivity level:** estimate of the impact of a failure of one of the security properties of a system low, medium or high

**service specific permissions:** field in an ETSI TS 103 097 [6] certificate (or, equivalently, an IEEE 1609.2 [i.26] certificate), bound to a specific ITS Application Identifier (ITS-AID), indicating the permissions of the certificate holder to carry out specific activities within the set of activities defined by that ITS-AID

**Time To Collision (TTC):** important calculated data element enabling the selection of the nature and urgency of the collision avoidance action to be undertaken

NOTE: This variable is related to other variables such as the road state, the weather conditions, the vehicle/VRU action capabilities. In the present document, TTC is used as a trigger for certain activities, such as starting to transmit VAMs or taking some avoidance/mitigation action. Research has been carried out on alternative metrics such as Post-Encroachment Time (PET) and Permutation Entropy (PE) (see [i.20] and [i.21]). The use of TTC in the present document is not meant to preclude the use of other metrics such as the triple of {Longitudinal Distance (LoD), Lateral Distance (LaD), Vertical Distance (VD)}, as elaborated in clause 6.5.10.5 that may be used to fulfil the safety goals of the applications, and in particular the present document does not prescribe a specific means for calculating TTC, specifically in order to support improvements in this algorithm in the future.

**Trajectory Interception Indication (TII):** indication of the likelihood that the VRU and one or more other VRUs, non-VRUs, or even objects on the road are going to collide (see also clauses 6.5.10.6 and 6.5.10.9)

**Vertical Distance (VD):** estimated distance in vertical direction (height) between the VRU and the vehicle

**velocity:** vector indicating speed in a particular direction (from ETSI EN 302 890-2 [5])

**Vulnerable Road Users (VRU):** non-motorized road users as well as users of VRU vehicles

NOTE: A VRU can only be living being. This living being is only considered as a VRU when it is in the context of a road safety related traffic environment.

**VRU application:** application extending the awareness of and/or about Vulnerable Road Users such as motorcycles, bicycles, pedestrians and impaired traffic participants in the neighbourhood of other traffic participants

**VRU cluster:** set of VRUs moving in a coherent manner, i.e. with coherent velocity or direction

NOTE: A cluster of VRUs can be homogeneous (set of pedestrians, of bicycles) or heterogeneous (pedestrians with e-scooters, bicycles). A combined VRU is not a heterogeneous cluster.

**VRU device:** portable device used by a VRU integrating a standard ITS station

NOTE: The definition of an ITS station is given in ETSI EN 302 665 [1]. A VRU device can also integrate applications interfacing the ITS-S. For example, an application can improve the VRU trajectory prediction by learning continuously from its behaviour when sharing the space with other road users.

**VRU device type:** class of VRU device transceiver capability, as either VRU-Tx, VRU-Rx or VRU-St (see clause 4.1)

**VRU ITS-S:** P-ITS-S/V-ITS-S capable of handling VRU related ITS applications

**VRU system:** ensemble of ITS stations interacting with each other to support VRU use cases, e.g. personal ITS-S, vehicle ITS-S, roadside ITS-S or Central ITS-S

**VRU vehicle:** L class of vehicles (for example mopeds or motorcycles, etc.), as defined in Annex I of EU Regulation 168/2013 [i.15] and light unpowered vehicles (bicycles, skates, wheelchairs, prams)

## 3.2 Symbols

Void.

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI EN 302 665 [1] and the following apply:

| | |
|---|---|
| AA | Authorization Authority |
| ADAS | Advanced Driver-Assistance Systems |
| AI | Artificial Intelligence |
| AT | Authorization Ticket |
| CA | Cooperative Awareness |
| CAM | Cooperative Awareness Message |

| | |
|---|---|
| CDD | Common Data Dictionary |
| C-ITS | Cooperative ITS |
| C-ITS-S | Central ITS Station |
| CPM | Collective Perception Message |
| CPS | Collective Perception Service |
| DCC | Decentralized Congestion Control |
| DDP | Device Data Provider |
| DE | Data Elements |
| DENM | Decentralized Environmental Notification Message |
| FAC | Facilities (layer) |
| FIPS | Federal Information Processing Standard |
| GNSS | Global Navigation Satellite System |
| HMI | Human-Machine Interface |
| ID | Identifier |
| IETF | Internet Engineering Task Force |
| IoT | Internet of Things |
| ITS | Intelligent Transport System |
| ITS-AID | ITS Application IDentifier |
| ITS-S | ITS Station |
| IVI | In Vehicle Information |
| LaD | Lateral Distance |
| LDM | Local Dynamic Map |
| LoD | Longitudinal Distance |
| MaaS | Mobility as a Service |
| MCM | Manoeuvre Coordination Message |
| MCS | Manoeuvre Coordination Service |
| MI | Manoeuvre Identifier |
| MSLaD | Minimum Safe Lateral Distance |
| MSLoD | Minimum Safe Longitudinal Distance |
| MSVD | Minimum Safe Vertical Distance |
| P-ITS-S | Personal ITS Station |
| POI | Point Of Interest |
| PoTi | Position and Time |
| PSM | Personal Safety Message |
| PTW | Powered Two-Wheelers |
| RFC | Request For Comments |
| RHS | Road Hazard Signalling |
| R-ITS-S | Roadside ITS Station |
| RSE | Roadside Equipment |
| RX | Receive |
| SAE | Society of Automotive Engineers |
| SPaT | Signal Phase and Timing |
| SSP | Service Specific Permissions |
| TDTC | Time Difference To Collision |
| TII | Trajectory Interception Indication |
| TLS | Transport Layer Security |
| TMC | Traffic Management Centre |
| TMS | Traffic Management System |
| TR | Technical Report |
| TS | Technical Specification |
| TTC | Time-To-Collision |
| TVRA | Threat, Vulnerability and Risk Analysis |
| TX | Transmit |
| UC | Use Case |
| V2I | Vehicle to Infrastructure |
| V2V | Vehicle to Vehicle |
| V2X | Vehicle-to-everything |
| VAM | VRU Awareness Message |
| VD | Vertical Distance |
| V-ITS-S | Vehicle ITS Station (VRU or non-VRU) |
| VRU | Vulnerable Road User |

# 4 General overview and use case analysis

## 4.1 Introduction

The present document specifies the requirements and the architecture of the VRU system and ITS Station (ITS-S), based on the results and analysis performed on exemplary use cases in ETSI TR 103 300-1 [i.1]. It describes the different functions involved in the identified architecture as well as their interfaces. Finally, it provides the impact of the VRU basic service introduction on other C-ITS standards and protocols.

A Vulnerable Road User (VRU) is considered to be a living being which participates in the road traffic. Such a living being is only in a role of a VRU when it acts in a safety related traffic context. As such, a VRU can also be assumed in other contexts, e.g. rail context.

Based on this definition, a bicycle or scooter itself is not considered as a VRU as long as no person uses this equipment (in this context, "use" means to deploy the equipment as intended). As long as the rider is on the bicycle, the combination (cluster) is a VRU, but as soon as the person is separated from the bicycle, the object "bicycle" is no longer a VRU. A standalone bicycle (i.e. not connected/controlled by a person) should not be protected as a VRU but signalled by a CPM or DENM as an object on the road. Transport devices which can become a VRU when used by a person in a safety related traffic context are called VRU vehicles.

Based on the analysis in ETSI TR 103 300-1 [i.1] the transmission of a VRU standard message is needed in a large majority of use cases. The present document introduces this VRU standard message, named VAM for VRU Awareness Message, designed as a separate message, as described in clause 6.9. The advantages of a VRU standard message are the following:

- a message different from the CAM message;
- a more flexible message in length and content;
- a message tentatively harmonised with the Personal Safety Message (PSM) defined in SAE J2945/9 [i.6].

As described in ETSI TR 103 300-1 [i.1], a VRU device can be of one of the three types defined in Table 1.

**Table 1: VRU device types**

| VRU device type | Description of VRU device |
|---|---|
| VRU-Tx | The device contains an ITS-S, having only a transmitter (no receiver) that broadcasts awareness messages or beacons about the VRU. The VRU device is however able to comply with Channel Congestion Control rules. |
| VRU-Rx | The device contains an ITS-S, having only a receiver and an HMI to receive messages from other ITS-S and can act upon the information received, e.g. inform or warn the VRU. |
| VRU-St | The device contains an ITS-S that includes the VRU-Tx and VRU-Rx functionalities. |

NOTE: In the present document, broadcast means a communication method used to transfer facilities layer messages to all endpoints.

The number of VRUs operating in a given area can get very high (see for example Use Case UC-B2 in [i.1], pedestrian crossing, where numbers can reach 500 or more) or the VRU can be combined with a VRU vehicle (e.g. rider on a bicycle). In order to reduce the amount of communication and thus the resource usage, VRU should be grouped together leading to VRU clusters which are indicated to other road users using a single VAM (called cluster VAM, see ETSI TS 103 300-3 [i.11]), rather than a VAM for each participant. These clusters can be homogeneous VRU clusters (for example group of pedestrians) or heterogeneous VRU clusters (e.g. made of different profiles of VRUs or constituting a combined VRU: rider on a bicycle). Each cluster is considered as a single entity and only the cluster leader will transmit the VAM. The parameters of the VRU cluster will be disseminated using the VAM.

## 4.2 Abstracted flow from use cases

The use cases in ETSI TR 103 300-1 [i.1] have shown the following steps (compiled as an abstraction of the use cases), with different alternatives. These steps were also highlighted by the VRUITS project [i.2]:

1) Detection of the VRU presence. The alternatives are:
   - VRU self-positioning. The VRU has sensors and potentially other sources allowing it to determine its own properties, including its location and velocity.
   - Another road user (e.g. a V-ITS-S) detects and tracks the VRU.
   - Roadside equipment connected to an R-ITS-S or a central ITS-S detects and tracks the VRU.
2) Evaluation whether the VRU is at potential risk from other road users and VRU position and dynamic state should be transmitted. Any party may transmit information about VRUs that it is aware of. Information on VRUs should be filtered and only be transmitted according to the message triggering conditions. The potential risk from other road users depends on the following conditions, among others:
   - Presence of other road users.
   - Road layout.
   - Dynamic state of the VRU and the other road users.
   - Traffic signal status for both VRU and vehicles, if relevant, and compliance to traffic lights.
3) Evaluation of safety message environment, specifically whether the VRU is part of a cluster, to determine whether the VRU's own ITS-S should transmit.
4) Transmission of information about VRU at-risk. Alternatives:
   - VRU sends ego-status information.
   - VRU cluster leader sends cluster information.
   - V-ITS-S, R-ITS-S, C-ITS-S or another road user sends information about a VRU in a potential risk situation.
5) Risk assessment. Phases (receiver side):
   - Fusion of sensor data, and observed information transmitted by other road users to build a local dynamic map, with information about road users' location, velocity and potentially other data, e.g. intention.
   - Assessment of risk based on estimated trajectory and velocity of different road users.
6) Warning or action to protect the VRU:
   - Warning of the device carrier (VRU or any other road user).
   - Transmission of collision warning to other road users.
   - Action in the case of an automated vehicle.

## 4.3 Messages for the use cases described in ETSI TR 103 300-1

This clause summarizes the C-ITS standardized messages that are involved in each of the use cases described in ETSI TR 103 300-1 [i.1]. The aim is to highlight the use cases defined in ETSI TR 103 300-1 [i.1] that require new or revised messages in ITS Facilities layer standards.

**Table 2: C-ITS messages for ETSI TR 103 300-1 [i.1] use cases**

| UC- | Description | Existing standard messages | | | | | | | Comments |
|---|---|---|---|---|---|---|---|---|---|
| | | CAM | DENM | SPaT | MAP | CPM | MCM | VAM | |
| A1 | Sharing sidewalk between pedestrian and cyclists | | X | | | | | X | VAM is used for awareness between VRUs, DENM is used for warning of a potential risk of collision (applies to all use cases below) |
| A2 | Pedestrian crossing a road with an e-scooter approaching | | X | | | | | X | |
| B1 | Active Roadwork | X | X | | | | | X | |
| B2 | VRU crossing a road | X | X | | | | X | X | |
| B3 | Rider is separated from his motorcycle | X | X | | | | X | | |
| B4 | Emergency Electronic Brake Light (EEBL) | X | X | | | | | | This UC is already covered by existing C ITS messages |
| B5 | Motorcycle Approach Indication (MAI) /Motorcycle Approach Warning (MAW) | X | | | | | | | CAM extended with complementary VAM information |
| C1 | Signalling VRU hidden by an obstacle | X | X | | | X | | X | |
| D1 | Signalled few VRUs in a protected area | X | X | | | X | X | X | |
| D2 | Non equipped VRUs crossing a road | X | X | | | X | X | | |
| D3 | VRUs crossing at a zebra protected by a traffic light | X | X | X | | X | X | X | |
| D4 | Scooter/bicyclist safety with turning vehicle | X | X | | | X | X | | VAM is not used because it is an unequipped VRU |
| E1 | Network assisted vulnerable pedestrian protection | X | X | | | X | X | X | |
| E2 | Detection of an animal or pedestrian on a highway | | X | | | X | X | X | |
| F1 | Signalled many VRUs in a protected area | X | X | | X | X | X | X | |
| F2 | Intelligent traffic lights for all (P2I2V) | X | | X | | | X | X | |
| NOTE: For UC-E2, the CAM is not used as it is not involved in the described use case. It could be present in an alternative use case when the central station disseminates a warning only in the case when it has detected an actual risk of collision, using the CAMs transmitted by the vehicles for this evaluation. | | | | | | | | | |

## 4.4 VRU system physical architecture

The physical architecture shall be as shown in Figure 1 (derived from VRUITS deliverable [i.2]). It consists of three main components:

- VRUs, including VRU clusters andcombined VRUs . The VRU ITS-S may be a personal ITS-S (P-ITS-S) or a vehicular ITS-S (V-ITS-S) depending on the type of road user.
- Non-VRU Vehicles (noted V), using V-ITS-S.
- Infrastructure (noted I), consisting of R-ITS-S or C-ITS-S.

NOTE: A motorcycle is considered as a VRU from VRU Profile 3 (see clause 6.1). As it is also considered as a non-VRU vehicle and transmits CAM messages, the present document recommends that a special vehicle container that identifies the motorcycle as VRU profile 3 (Motorcyclist special container, with complementary data elements) be included in the CAM message.

As shown in Figure 2, the overall environment comprises ITS stations (ITS-S) that may communicate directly as follows:

- from VRU to Vehicle and vice versa (VRU2V/V2VRU)
- from VRU to VRU (VRU2VRU)
- from VRU to Infrastructure and vice versa (VRU2I/I2VRU)
- from Vehicle to Vehicle (V2V)
- from Vehicle to Infrastructure and vice versa (V2I/I2V)

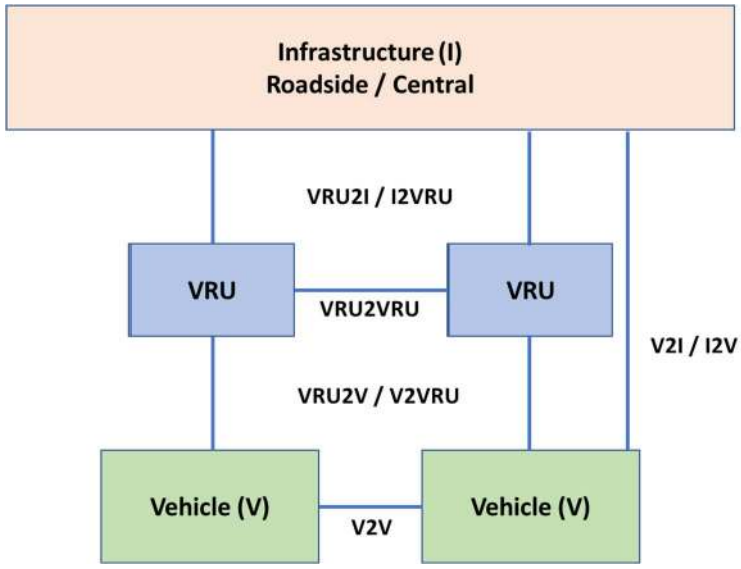



**Figure 1: VRU system physical architecture (adapted from VRUITS project [i.2])**

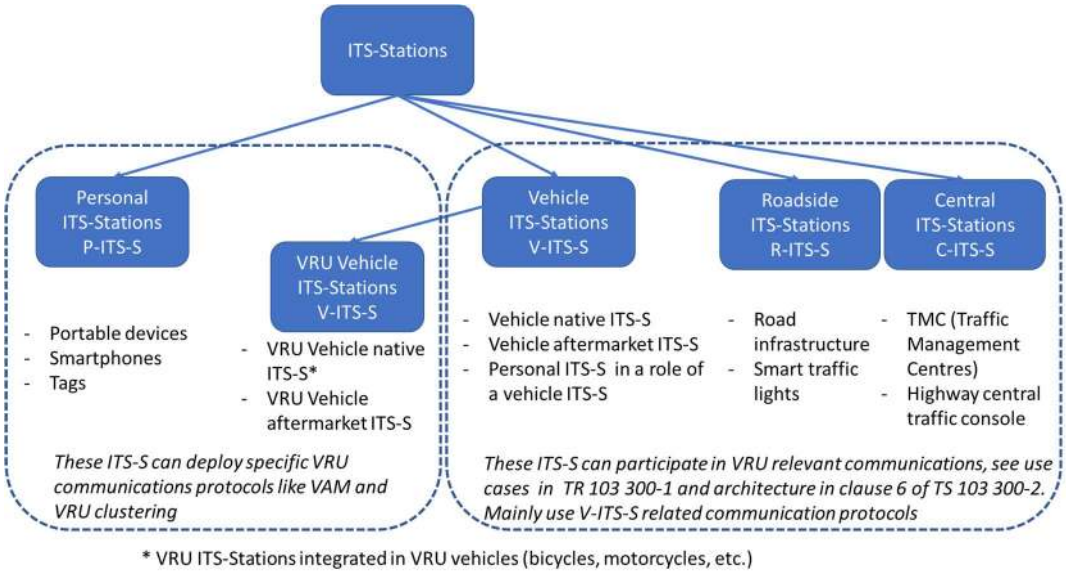



**Figure 2: ITS-Stations (ITS-S) in the VRU system**

Figure 2 illustrates the notion of VRU ITS-S (left pane) and non-VRU ITS-S (right pane).

The arrows in the diagram indicate sub-classes. VRU vehicle ITS-S is a sub-class of V-ITS-S, as shown by the arrows in Figure 2.

NOTE 1: A P-ITS-S can take the role of a VRU V-ITS-S, e.g. when connected to a VRU vehicle such as a bicycle or a motorcycle. In this case it can deploy VRU protocols sending VAMs with a different profile, e.g. in the case of a combined VRU composed of a VRU and a VRU vehicle. This may also happen when there is no VRU device in the VRU vehicle.

NOTE 2: It is necessary to distinguish between the physical implementation (P-ITS-S, V-ITS-S, R-ITS-S and C-ITS-S) and the role of the communication mechanisms deployed. Role and communication mechanisms can be deployed synchronously, e.g. a V-ITS-S can send out vehicular related messages or VRU related messages depending on its role.

NOTE 3: A physical implementation of a VRU-only ITS-S is possible. This is not a full P-ITS-S but rather a subclass of P-ITS-S. This station can only deploy VRU related protocols, it might be a TX-only device (but still complying with channel congestion control rules) or a device with limited capabilities. This might be needed to reduce power consumption and costs. These devices might not be capable of taking the role of a VRU cluster leader.

## 4.5 Security analysis of VRU use cases

### 4.5.0 Introduction

This clause presents the security analysis of the VRU use cases considered in ETSI TR 103 300-1 [i.1].

This security analysis considers each "entity activity" (i.e. each information flow with a particular goal) in the use cases. Each entity activity is rated as low, medium or high sensitivity with respect to the three security properties confidentiality (C), integrity (I) and availability (A) as identified in ISO/IEC 27001 [i.16] and Federal Information Processing Standard (FIPS) 199 [i.17]:

- The confidentiality sensitivity measures how severe the effect is, if information is read by a party that should not read it.

- The integrity sensitivity measures how severe the effect is if information is trusted by a party when the information is incorrect. This concept captures both intentionally false data, introduced by an attacker, and data that is honestly inaccurate without the receiver knowing it.

NOTE: This meaning of "integrity" is different from its meaning in a purely cryptographic context - in that context the term simply means assurance that data has not been modified since it was created by a legitimate party and does not include considerations of data quality at the time of creation.

  The integrity sensitivity level takes into account how the receiver is expected to behave on receipt of information, as this affects the impact of an integrity failure.

- The availability sensitivity measures how severe the effect is if information is not received by the party that relies on receiving it.

Sensitivity levels are categorized as low, medium or high: low indicates a limited adverse effect, medium indicates a serious adverse effect, and high indicates a critical or catastrophic effect. Confidentiality can also have sensitivity level of "none", indicating that the data is public.

As a guide to how to think about sensitivity levels, examples of possible adverse outcomes and the corresponding sensitivity levels are:

- Events that can cause false collision warnings are categorized as having integrity sensitivity "Low". These events are undesirable and should be mitigated but are extremely unlikely to lead to physical harm or to significant financial losses.

- Events that could cause a single collision are categorized as having integrity sensitivity "Medium" (no such events are identified in the analysis below). This indicates that the outcome is extremely undesirable and should be prevented but that in the big picture it is possible to accept a non-zero number of these events.

- Events that could cause widespread collisions from a single event are categorized as having integrity sensitivity "High" (no such events are identified in the analysis below). This indicates that the outcome is so severe that the system should not be deployed unless it can be guaranteed that no such false events will occur.

As automated vehicles become more widespread, there will be an increased potential that "low" or "medium" sensitivity events can be scalable to create widespread incidents. The analysis performed in the present document does not reflect this potential scalability and focuses on messages used to raise warnings to human drivers. The assumption is that autonomous drivers will go through a process of fusion and decision similar to human drivers and will ultimately behave similarly to human drivers from the point of view of security. Cyber-vulnerabilities due to bad implementations of autonomous systems are not considered in the present document as they are unpredictable and may have an impact that is unconnected with the initial use case, making analysis purely speculative.

The C/I/A analysis enables the derivation of security requirements.

One set of security requirements is derived directly from the sensitivity analysis: for any information flow, communications security mechanisms should be specified to meet the sensitivity requirements for confidentiality (provided by encryption) and integrity (provided by authentication) and communications assurance mechanisms should be specified to meet the sensitivity requirements for availability.

Additionally, the sensitivity requirements of the information flows involving a particular actor within the use case indicate security controls that the actor should implement. Those security controls can be physical, such as the inclusion of hardware security modules, and/or process or organizational, such as following a data management plan for secure storage and deletion of generated data or specifying particular methods to be used to determine that an actor is entitled to certificates with a particular set of permissions. These process controls can also include risk reduction measures taken on receipt of a message.

Within the set of integrity controls, one control that can be used to manage risk is separation of roles within the application. In this context, "separation of roles" means "identifying different groups of activities within the application such that an actor in one role can carry out activities from only some of those groups". If separate roles are needed, they will typically be addressed by defining a Service Specific Permissions (SSP) structure for the application. This matches the methodology recommended in SAE J2945/5 [i.3].

Separately, the security analysis for the information flows should consider requirements for pseudonymity. Pseudonymity is in general required when devices operated by or associated with private citizens send messages with no confidentiality mechanisms applied. The standard pseudonymity mechanism is to provide those devices with multiple digital certificates, allowing the device to use different certificates to sign different messages and so inhibiting the ability of an eavesdropper to determine whether two messages sent at different times and in different places have come from the same device. Pseudonymity requirements are captured in the notes for the security analysis for each use case.

The present document does not present a full Threat, Vulnerability and Risk Analysis (TVRA) carried out per ETSI TS 102 165-1 [i.7] as there are elements of the full TVRA process, such as estimating the attack likelihood, that are impossible for a system that has not yet been deployed and may depend on specifics of implementation.

The present document also does not consider requirements for personal data protection once the data has been transmitted and used for its immediate purpose of VRU protection. Per the draft guidelines of the European Data Protection Board on processing personal data in the context of connected vehicles [i.22], this data should only be used for the intended purpose. However, the present document does not rule out the possibility that in future there may be additional services, provided in the interest of the user generating the data and intended by the user, which require the retention of the data beyond its immediate use for safety of life protection. The present document therefore does not require any specific data protection or retention policies for received data. Implementers and deployers should refer to the current European Data Protection Board guidance to understand requirements on data protection and retention for VRU applications in the EDPB area.

### 4.5.1 UC-A1: Sharing sidewalk between pedestrian and cyclists

**Table 3: Risk analysis: UC-A1: Sharing sidewalk between pedestrian and cyclists**

| Use case: Sharing sidewalk between pedestrian and cyclists | | | |
|---|---|---|---|
| Actors: Pedestrian VRU, Cyclist VRU | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IA1-1 Exchange messages | EA1-1.1 VRU → VRU | Broadcast of dynamic and other information about sending VRU | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in the use case require pseudonymity. | | | |

### 4.5.2 UC-A2: Pedestrian crossing a road with an e-scooter approaching

**Table 4: Risk analysis: UC-A2: Pedestrian crossing a road with an e-scooter approaching**

| Use case: Pedestrian crossing a road with an e-scooter approaching | | | |
|---|---|---|---|
| Actors: Pedestrian VRU, Cyclist VRU | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IA2-1 Exchange messages | EA2-1.1 E-Scooter → Pedestrian | Broadcast of dynamic and other information about E-Scooter | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EA2-1.2 Pedestrian → E-Scooter | Broadcast of dynamic and other information about pedestrian | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| IA2-2 Warn other vehicles | EA2-2.1 Pedestrian notifies passing-by vehicles | Warning of potential risk that E-Scooter will create hazard | C: None - broadcast model.<br>I: Low - False warning makes drivers drive more carefully, but they should be driving carefully in this environment anyway.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in the use case require pseudonymity. | | | |

### 4.5.3 UC-B1: Active roadwork

**Table 5: Risk analysis: UC-B1: Active roadwork**

| Use case: Active roadwork | | | |
|---|---|---|---|
| Actors: Active worker (pedestrian) VRU, Local Vehicle, Remote Vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IB1-1 Exchange messages | EB1-1.1 Local Vehicle → Active Worker | Broadcast of dynamic and other information about local vehicle | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB1-1.2 Active Worker → Local Vehicle | Broadcast of dynamic and other information about active worker | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB1-1.3 Local Vehicle → Remote Vehicle | Broadcast of notification about presence of active worker | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB1-1.4 Remote Vehicle → Local Vehicle | Broadcast of dynamic and other information about remote vehicle | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB1-1.5 Local Vehicle → Active Worker | Broadcast of notification about presence of hazardous vehicle | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in the use case require pseudonymity. | | | |

### 4.5.4 UC-B2: VRU crossing a road

**Table 6: Risk analysis: UC-B2: VRU crossing a road**

| Use case: VRU crossing a road | | | |
|---|---|---|---|
| Actors: VRU, Vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IB2-1 Exchange messages | EB2-1.1 VRU → Vehicle | Broadcast of dynamic and other information about VRU | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB2-1.2 Vehicle → VRU | Broadcast of dynamic and other information about vehicle | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB2-1.3 Vehicle → VRU | Warning of collision risk | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in the use case require pseudonymity. | | | |

### 4.5.5 UC-B3: Rider is separated from his motorcycle

**Table 7: Risk analysis: UC-B3: Rider is separated from his motorcycle**

| Use case: Rider separated from motorcycle | | | |
|---|---|---|---|
| Actors: Rider VRU-St, motorcycle VRU-St, Approaching Vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IB3-1 Exchange messages | EB3-1.1 Rider VRU-St → Motorcycle VRU-St: communications to establish whether VRU is on board bicycle, (see note 2) | Information used by VRU device to determine whether VRU is still on motorcycle | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB3-1.2 Motorcycle VRU-St → Vehicle | Message warning of separated combined VRU | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB3-1.3 Rider VRU-St → Vehicle | Message warning of separated combined VRU | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in information flow EB3-1.2 and EB3-1.3 require pseudonymity. In information flow EB3-1.1 the actors do not necessarily require pseudonymity from each other but the mechanism used to implement information flow EB3-1.1 should provide pseudonymity from other road users.<br>NOTE 1: The security mechanism to be used for EB3-1.1 depends on the specific design of that information flow.<br>NOTE 2: The exact mechanism used is not specified in the present document and, while it may be based on VAMs from the motorcycle, it may also be sensor based or based on non-C-ITS communications. | | | |

### 4.5.6 UC-B4: Emergency Electronic Brake Light (EEBL)

**Table 8: Risk analysis: UC-B4: Emergency electronic brake light**

| Use case: Emergency Electronic Brake Light (EEBL) | | | |
|---|---|---|---|
| Actors: Braking ITS-S, Receiving ITS-S | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IB4-1 Exchange messages | EB4-1.1 Braking ITS-S → Receiving ITS-S | Broadcast of information about brake event | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in the use case require pseudonymity. | | | |

### 4.5.7 UC-B5: Motorcycle Approach Indication (MAI)/Motorcycle Approach Warning (MAW)

**Table 9: Risk analysis: UC-B5: Motorcycle approach indication/motorcycle approach warning**

| Use case: Motorcycle approach indication/motorcycle approach warning | | | |
|---|---|---|---|
| Actors: Motorcycle ITS-S, Vehicle ITS-S, Receiving ITS-S (which may be motorcycle or vehicle) | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IB5-1 Exchange messages | EB5-1.1 Motorcycle ITS-S → Receiving ITS-S | Broadcast of dynamic information about the VRU | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EB5-1.2 Vehicle ITS-S → Motorcycle ITS-S | Broadcast of dynamic and other information about vehicle | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in the use case require pseudonymity. | | | |

### 4.5.8 UC-C1: Signalling VRU hidden by an obstacle

**Table 10: Risk analysis: UC-C1: Signalling VRU hidden by an obstacle**

| Use case: Signalling VRU hidden by an obstacle | | | |
|---|---|---|---|
| Actors: VRU, Sensor-equipped vehicle, Receiving vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IC1-1 Detect | EC1-1.1 Sensor-equipped vehicle detecting VRU | Sensor activity to detect VRU | C: None - VRU is detectable by any sensor-equipped vehicle.<br>I: Low - Impact of false sensor data is false warnings.<br>A: Low - Any level of availability allows this use case to be effective. |
| IC1-2 Warn | EC1-2.1 Sensor-equipped vehicle → Receiving vehicle | Perception message | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in the use case require pseudonymity.<br>NOTE 1: The integrity requirement on EC1-1.1 reflects the requirement for integrity as applied to sensor data fed to the V-ITS-S on the sensor-equipped vehicle.<br>NOTE 2: Although the impact of an integrity failure is the same in this case (sensed VRU) as in other cases (self-reporting VRU), it may be appropriate to define an SSP-protected role for the sensor-equipped vehicle to provide assurance that vehicles that send the perception message have in fact been certified as being equipped with the appropriate sensors. | | | |

### 4.5.9 UC-D1: Signalled few VRUs in a protected area

**Table 11: Risk analysis: UC-D1: Signalled few VRUs in a protected area**

| Use case: Signalled few VRUs in a protected area | | | |
|---|---|---|---|
| Actors: VRU, RSE, Approaching vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| ID1-1 Detect | ED1-1.1 Approaching vehicle → RSE | Broadcast of dynamic and other information about vehicle used to predict hazard situation | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| ID1-2 Warn | ED1-2.1 RSE → Approaching vehicle | Broadcast of warning about protected area | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | ED1-2.2 RSE → VRUs | Warning Siren | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: All actors in the use case except for the RSE require pseudonymity.<br>NOTE: A false CAM in information flow ED1-1.1 may cause a false alarm. The result of a false alarm is an evacuation of the protected area. If the protected area is a work zone, this may result in damage to equipment being used in the protected area and prolonged disruption to work done in the protected area. Similarly, even if the protected area is not a work zone, it is possible (though not high probability) that VRUs will be injured during the evacuation. However, this still does not meet the threshold for Medium integrity sensitivity as it does not cause disruption, or a threat of physical damage, on a sufficiently large scale or with sufficiently high probability. | | | |

### 4.5.10 UC-D2: Non equipped VRUs crossing a road

**Table 12: Risk analysis: UC-D2: Non equipped VRUs crossing a road**

| Use case: Non equipped VRUs crossing a road | | | |
|---|---|---|---|
| Actors: Notifying ITS-S, Receiving vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| ID2-1 Warn | ED2-1.1 RSE → Approaching vehicle | Broadcast of warning information of VRUs in road | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: The RSE is the only transmitter in this case and does not require pseudonymity.<br>NOTE 1: This description omits the CAM used by the RSE to detect vehicles as this activity will happen anyway, outside the context of the VRU use case.<br>NOTE 2: The traffic signal RSE in this case is assumed to be equipped with more sensors than a traffic signal RSE that simply sends SPaTs. This therefore creates a potential requirement for a role identifier (an SSP) that distinguishes between sensor-equipped RSEs and non-sensor-equipped RSEs. However, since the impact of a false message is low, the recommendation of this analysis is that a separate role identifier is not necessary. | | | |

### 4.5.11 UC-D3: VRUs crossing at a zebra protected by a traffic light

**Table 13: Risk Analysis: UC-D3: VRUs crossing at a zebra protected by a traffic light**

| Use case: VRUs crossing at a zebra protected by a traffic light in the presence of a priority vehicle | | | |
|---|---|---|---|
| Actors: Priority vehicle, RSE, VRU | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| ID3-1 Request priority | ED3-1.1 Priority vehicle → RSE | message implicitly or explicitly requesting signal prioritization | C: None - broadcast model.<br>I: Medium - False signal prioritization messages can cause significant traffic disruption.<br>A: Low - Any level of availability allows this use case to be effective. |
| ID3-2 Detect VRUs | ED3-2.1 Sensor-equipped RSE detecting VRU | Sensor activity to detect VRU | C: None - VRU is detectable by any sensor-equipped vehicle.<br>I: Low - The worst outcome is false warnings but these will be delivered to a series of different drivers and so will not significantly affect the confidence of any driver in the system.<br>A: Low - Any level of availability allows this use case to be effective. |
| ID3-3 Send Signal Prioritization Result | ED3-3.1 RSE → Priority vehicle | Signal priority request result: message indicating presence of VRUs, or indicating that priority will (not) be granted | C: None - VRU is detectable by any sensor-equipped vehicle.<br>I: Low -Priority vehicle should be moving with caution anyway since it is approaching a signalized intersection and so a false safety indication should not result in excessive speed.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: Neither the RSE nor the priority vehicle require pseudonymity.<br>NOTE 1: The traffic signal RSE in this case is assumed to be equipped with more sensors than a traffic signal RSE that simply sends SPaTs. This therefore creates a potential requirement for a role identifier (an SSP) that distinguishes between sensor-equipped RSEs and non-sensor-equipped RSEs. However, since the impact of a false message is low, the recommendation of this analysis is that a separate role identifier is not necessary.<br>NOTE 2: The design of the traffic signal prioritization mechanism in ED3-1.1 is not in the scope of the present document and the security analysis is included only for completeness. | | | |

### 4.5.12 UC-D4: Scooter/bicyclist safety with turning vehicle

**Table 14: Risk analysis: UC-D4: Scooter/bicyclist safety with turning vehicle**

| Use case: Scoots/bicyclist safety with turning vehicle | | | |
|---|---|---|---|
| Actors: VRU, RSE, vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| ID4-1 Detect | ED4-1.1 Sensor-equipped RSE detecting VRU | Sensor activity to detect VRU | C: None - VRU is detectable by any sensor-equipped vehicle.<br>I: Low - False positives will affect a number of different drivers and will not have a significant impact on confidence in the system.<br>A: Low - Any level of availability allows this use case to be effective. |
| ID4-2 Warn | ED4-2.1 Sensor-equipped RSE → Vehicle | Perception message | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity requirements: The vehicle requires pseudonymity, the RSE does not.<br>NOTE: The traffic signal RSE in this case is assumed to be equipped with more sensors than a traffic signal RSE that simply sends SPaTs. This therefore creates a potential requirement for a role identifier (an SSP) that distinguishes between sensor-equipped RSEs and non-sensor-equipped RSEs. However, since the impact of a false message is low, the recommendation of this analysis is that a separate role identifier is not necessary. | | | |

### 4.5.13 UC-E1: Network assisted vulnerable pedestrian protection

**Table 15: Risk analysis: UC-E1: Network assisted vulnerable pedestrian protection**

| Use case: Network assisted vulnerable pedestrian protection | | | |
|---|---|---|---|
| Actors: VRU, vehicle, third party in cloud | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IE1-1 Detect | EE1-1.1 VRU → Third party in cloud | Position report | C: None - VRU is detectable by any sensor-equipped vehicle.<br>I: Medium - If location can be spoofed then attacks will be scalable.<br>A: Low - Any level of availability allows this use case to be effective. |
| IE1-2 Warn | EE1-2.1 Third party in cloud → Vehicle | Perception message | C: None - broadcast model.<br>I: Medium - If this communications session can be hacked then the system is vulnerable to widespread false alerts and disruption.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity: The VRU may be identified to the third party in the cloud if the VRU has given consent, if the third party follows good privacy and data management practices, and if the third party does not reveal the identity of the VRU to the third party. The third party in the cloud does not need pseudonymity. The Vehicle does not send messages in this use case.<br>NOTE: The security mechanisms on the two information flows in this scenario need not be based on the current (role-based) ETSI ITS security specifications but may use existing (identity-based) web authentication technologies. | | | |

### 4.5.14 UC-E2: Detection of an animal or pedestrian on a highway

**Table 16: Risk analysis: UC-E2: Detection of an animal or pedestrian on a highway**

| Use case: Network assisted vulnerable pedestrian protection | | | |
|---|---|---|---|
| Actors: Roadside Camera, vehicle, third party in cloud, VRU | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IE2-1 Detect | EE2-1.1 Roadside camera → Control centre | VRU report | C: None - VRU is detectable by any sensor-equipped vehicle.<br>I: Medium - If location can be spoofed then attacks will be scalable.<br>A: Low - Any level of availability allows this use case to be effective. |
| IE2-2 Warn | EE2-2.1 Third party in cloud → Vehicle | Broadcast of notification about presence of hazardous VRU | C: None - broadcast model.<br>I: Medium - If this communications session can be hacked then the system is vulnerable to widespread false alerts and disruption.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EE2-2.2 Third party in cloud → Equipped VRU | Broadcast of warning message | C: None - broadcast model.<br>I: Low - A false warning to a VRU will not cause significant widespread disruption to the system.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity: None of the sending parties in this use case need pseudonymity.<br>NOTE: The security mechanisms on the two information flows in this scenario need not be based on the current (role-based) ETSI ITS security specifications but may use existing (identity-based) web authentication technologies. | | | |

### 4.5.15 UC-F1: Signalled many VRUs in a protected area

**Table 17: Risk analysis: UC-F1: Signalled many VRUs in a protected area**

| Use case: Signalled many VRUs in a protected area | | | |
|---|---|---|---|
| Actors: Equipped VRUs, RSE, Approaching vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IF1-1 Detect | EF1-1.1 Approaching vehicle → RSE | Broadcast of dynamic and other information about vehicle | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EF1-1.2 Equipped VRU → RSE | Broadcast of dynamic and other information about VRUs | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| IF1-2 Warn | EF1-2.1 RSE → Approaching vehicle | Broadcast of warning of protected area | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EF1-2.2 RSE → VRUs | Warning Siren or broadcast of notification about presence of vehicle | C: None - broadcast model.<br>I: Low - Impact of false messages is false alarm for individuals.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity: The VRUs and the vehicles need pseudonymity, the RSE does not.<br>NOTE: Even though a false alarm would cause disruption to a potentially large number of VRUs, this is still not a sufficiently large disruption to count as Medium integrity sensitivity. | | | |

### 4.5.16 UC-F2: Intelligent traffic lights for all

**Table 18: Risk analysis: UC-F2: Intelligent traffic lights for all**

| Use case: Intelligent traffic lights for all | | | |
|---|---|---|---|
| Actors: Privileged VRU, RSE, Vehicle | | | |
| **Information flow** | **Entity activities** | **Description** | **Impact** |
| IF2-1 Detect | EF2-1.1 Equipped VRU → RSE | Awareness message of<br>(a) presence of VRU<br>(b) fact that VRU is privileged<br>(c) fact that VRU is about to cross or is crossing the road | C: None - broadcast model.<br>I: Low - If there are bad actors who falsely claim to be privileged VRUs, the system can still manage the impact by asking them to wait before crossing the road, etc.<br>A: Low - Any level of availability allows this use case to be effective. |
| IF2-2 Extended signal notification | EF2-2.1 RSE → Approaching vehicle | Indication that red will be extended | C: None - broadcast model.<br>I: Low - Message should be in synch with traffic signals and is for information only.<br>A: Low - Any level of availability allows this use case to be effective. |
| | EF2-2.2 RSE → VRUs | Indication that red signal will be extended | C: None - broadcast model.<br>I: Low - Message is unlikely to affect VRU behaviour in a way that has wider impact on the traffic system.<br>A: Low - Any level of availability allows this use case to be effective. |
| Pseudonymity: The VRU and the vehicle need pseudonymity; the RSE does not.<br>NOTE: The impact of an integrity breach on EF2-1.1 is low, because excessive requests can be handled by adjusting the vehicle movement phase length or by choosing not to prolong the crossing phase immediately after a request instead interspersing prolonged phases between normal-length phases. As such, although these "social weak" VRUs have elevated privileges compared to ordinary VRUs, it is not clear that there is a security requirement for the "social weak" VRU role to be separated from the ordinary VRU role. The present document nevertheless recommends that a "social weak" VRU role is identified to help manage the risk of potential fake social weak VRUs in this and other contexts. | | | |

# 5 VRU related requirements

## 5.1 Introduction

This clause specifies the requirements on the VRU-related components in the C-ITS. These requirements are derived from the use cases and related analysis described in clause 6 and in clause 7 of ETSI TR 103 300-1 [i.1].

The requirements are classified into two categories: functional requirements, which will have an impact on the VRU system architecture and operational requirements to be validated when the VRU system will be tested and deployed. The so identified requirements are further considered at system, communication and security level. Some of these requirements may feed into further work, such as the communication architecture to be used by the VRU system (see ETSI TR 103 300-1 [i.1], clause 4.3 for the definition of the VRU system).

The requirements have been numbered using the following naming convention: XYYYnn.i, where:

- X= F for functional/O for operational
- YYY = SYS for system, COM for communication, SEC for security
- nn is a sequential number identifying the requirement

- i is an optional sequential number identifying sub-requirements when different options exist on a specific requirement.

NOTE: Requirements at station level are included in the system level as ITS-S are part of the C-ITS.

## 5.2 Functional requirements

### 5.2.1 System requirements

System functional requirements identify the ITS functions necessary to support the VRU services.

**Table 19: Functional system requirements**

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| FSYS01 | An element of the VRU system shall use an ITS-S (either personal, vehicle, roadside or central) for transmission and shall either be integrated with that ITS-S or have a secure connection to that ITS-S. | The VRU system is made of a composition of interactive elements selected among the four following elements: VRU device(s), non-VRU vehicle(s), roadside equipment, central system(s).<br>Each constituting element integrates one ITS station. | [i.1], clause 6. |
| FSYS02 | The VRU system shall include functions aiming at detecting a risk of collision between VRUs and other road users and avoiding such collisions. | These functions and associated data can be distributed in an optimal way in different elements of the VRU system. | [i.1], clause 6. |
| FSYS03 | The VRU system architecture shall support several possible interaction scenarios between the four elements of ITS architecture it is made of. | These interaction scenarios depend on the distribution of functions and data in the four possible types of ITS stations (personal device, vehicle, RSE, central) of the VRU system. | [i.1], clauses 6 and 7. |
| FSYS04 | The VRU system shall be able to fuse incoming C-ITS messages with other data, if available, to form a dynamically updated map of the motion of relevant road users. | The collision risk analysis function is one key functional element which relies on the ITS elements to be able to fully and constantly perceive the movements of mobiles which are possibly colliding. | [i.1], clauses 6, 7.2, 7.8, 7.9 and 7.12. |
| FSYS05 | Calculations of risk to the VRU shall be based on all relevant physical properties, including but not necessarily limited to the dynamic state and the mass of the moving objects. | The collision risk analysis function considers the full dynamic state of the road users and predicts their respective changes. This may be used to determine the most appropriate collision avoidance strategy and action. | [i.1], clauses 6, 7.2, 7.8, 7.9 and 7.12. |
| FSYS06 | The VRU collision risk analysis function, and dynamic state prediction shall be able to reliably predict the relevant road users manoeuvres with an acceptable level of confidence for the purpose of triggering the appropriate collision avoidance action, assuming that the input data is of sufficient quality. | The collision risk analysis function analyses the level of collision risk based on a reliable prediction of the respective dynamic state evolution. Consequently, the reliability level aspect needs to be characterized in terms of confidence level for the chosen collision risk metrics as discussed in clauses 6.5.10.5 and 6.5.10.9. | [i.1], clauses 7.2, 7.8, 7.9 and 7.12. |
| FSYS07 | The collision avoidance actions shall be triggered at the level of the VRU or the vehicle or both, depending on the VRU capabilities to act (VRU profile and sub-profile), the vehicle type and capabilities and the actual risk of collision. | VRU do not always have the capability to act to avoid a collision (ex: animal, children, aging person, disabled, etc.), especially if the TTC is short (a few seconds). See clauses 6.5.10.5 and 6.5.10.6. | [i.1], clauses 7.2, 7.9 and 7.12. |

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| FSYS08 | Depending on the vehicle level of automation, the collision avoidance action or impact mitigation action shall be triggered as a warning/alert to the driver or as a direct action on the vehicle itself. | The following collision avoidance actions can be envisioned at this point (may be combined):<br>• extend or change the phase of a traffic light;<br>• act on the trajectory and/or velocity of the vehicle (slow down, change lane) if the vehicle has a sufficient level of automation;<br>• alert the ITS device user through the HMI (depends on device capabilities and VRU profile and sub-profile, see FSYS14);<br>• disseminate a C-ITS message to other road users, including the VRU if relevant.<br>The following impact mitigation actions can be envisioned at this point (may be combined):<br>• trigger a protective mechanism at the vehicle level (e.g. airbag);<br>• trigger a portable VRU protection airbag. | [i.1], clause 7.8 and use case B1. |
| FSYS09 | At the VRU level, the reliable prediction of the VRU movement shall take into account the VRU profile (see clause 6.1) with the purpose to provide a reasonable level of confidence in this prediction. | The VRU dynamic state prediction algorithm may be improved over time by local learning about the usual behaviour of specific VRUs or by other means. | [i.1], clauses 7.2, 7.4 and 7.5. |
| FSYS10 | The VRU basic service of a VRU ITS-S shall be deactivated as long as the VRU device owner is inside a non-VRU vehicle (e.g. a vehicle with a strong exterior) or protected area and cannot be considered any more as vulnerable. The VRU basic service shall be activated after the VRU device owner leaves the protected vehicle. | A Road user may not be vulnerable anymore when entering a protected area or using a non VRU transportation means. Consequently, the VRU role transition is managed with the objective to adapt the VRU service, even deactivate it when not necessary anymore, as soon as the VRU status disappears. The VRU service remains operational in both conditions. | [i.1], clause 7.2. |
| FSYS11 | The configuration of all elements of the VRU system, including changes and updates that may be necessary during their life cycles, shall be managed. | Configuration management is necessary for the update of the ITS station during its life cycle. It is necessary for preventive and corrective maintenance, but also for technological, protocol evolutions and more generally functional evolutions. | [i.1], clause 7.7. |
| FSYS12 | When participating to the VRU system, a R-ITS-S associated to a traffic light shall be able to adapt the phase duration according to the VRU profile and sub-profile, or send warnings to provide additional safety for VRUs. | A Roadside Equipment that can only send warnings is still useful, even if it cannot affect signal phase and timing. | |
| FSYS13 | VRU devices involved in C-ITS communications shall support at least one of the following configurations (VRU device types): VRU-Tx, VRU-Rx, VRU-St, as defined in clause 4.1. | A VRU device configured as a VRU-Tx or a VRU-St complies with the channel congestion control rules as defined in ETSI TS 103 175 [i.18]. | [i.1], use case A1. |
| FSYS14 | VRU devices supporting VRU-Rx/VRU-St configuration shall include an appropriate HMI to notify their users in case of safety issue. | This HMI can be audio, text, graphics or any means relevant with the VRU profile. | [i.1], use case A1. |

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| FSYS15 | When relevant, a VRU device shall indicate special VRU conditions such as very vulnerable person, disabled, pram, wheelchair, etc. | This indication is optional, but will help for example traffic lights adapt their phase accordingly. This requirement may be difficult to assert as personal devices maybe exchanged between users, for example a disabled person lending his personal device to a child. | [i.1], use case E1. |

### 5.2.2 Communication requirements

**Table 20: Functional communication requirements**

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| FCOM01 | A VRU system shall operate effectively when up to [5 000] number of users are within the same communications zone, i.e. within a circle of radius up to 300 m, as defined in the Road Hazard Signalling (RHS) standard [7]. This may be achieved by the clustering of active users. | The maximum number of users depends on the density of VRUs in a geographical area. Moreover, this also depends on standardized communication protocols including channel congestion control rules and on the clustering efficiency.<br>EXAMPLE: Considering a crowd of VRUs within a communication distance of 100 meters, in a quadratic area of 10 000 $m^2$, which can be divided in 1 000 bounding boxes of 10 $m^2$ each. This results in 1 000 VRU clusters, i.e. 1 000 users of the VRU system. | [i.1], use case F1. |
| FCOM02 | A VRU system shall support an appropriate congestion control mechanism. | Channel congestion control rules consider the maximum number of VRUs which may be communicating simultaneously as well as the available channel bandwidth. | [i.1],clause 7.5. |
| FCOM03 | VRU devices shall operate to mitigate a channel congestion problem if there are many ITS-S in the VRU environment. | Possible solutions for prevention and reduction of the number of transmitted messages are:<br>• Consulting appropriate maps to verify if the VRU is on a drivable road or in protected areas such as buildings before sending VAMs.<br>• Designating a geographical area as a pedestrian zone (using the MAPEM).<br>• Considering itself as part of a VRU cluster when relevant and when transmitting VAM.<br>• Considering the VRUs as part of a VRU cluster when transmitting CPM by V-ITS-S or R-ITS-S.<br>• Considering CPMs from other sources to detect if the VRU has already been reported by others. | [i.1], use cases A1, A2, B1, B2, B5, F1. |
| FCOM04 | A VRU system shall support the flexible and dynamic triggering of messages with generation intervals from 100 ms at the most frequent. | The VAMs frequency is related to the VRU motion dynamics and chosen collision risk metric as discussed in clause 6.5.10.5. | |

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| FCOM05 | A VRU device supporting the VRU-Tx or VRU-St configuration shall be able to broadcast VAMs, with sufficient frequency to enable timely detection of a collision risk. | See clause 4.3. | [i.1], use cases A1, A2, B1, B2, B3, B5, F1, F2 and clause 7.5. |
| FCOM06 | When the VRU ITS-S is able to transmit (VRU-Tx or VRU-St configuration), the VRU ITS-S shall adapt the periodicity of transmitted VAMs to its profile, velocity, direction, context, being part or not of a VRU cluster, and potentially to the evaluation of increased risk. | | [i.1], use cases A1, A2, B1, B2, B3, B5, F1, F2 and clause 7.5. |
| FCOM07 | The VAM shall include in its data elements the VRU position, its dynamic state, its dimensions, the current VRU profile (as defined in clause 6.1) and whether it is part of a cluster of VRUs.<br>The VAM may include the mass and size class of the VRU for a combined VRU (e.g. a bicyclist) and the trajectory of the VRU including the history and predicted future trajectory | These data elements are necessary to the collision risk analysis. | [i.1], use case B3. |
| FCOM08 | When relevant, V-ITS-S, R-ITS-S and C-ITS-S shall transmit VRU related warning and awareness messages selecting the most appropriate message. | DENM and CPM have been considered in the present document. See note. | [i.1], use case C1. |
| FCOM09 | A VRU-related warning or awareness message shall contain at least the following data elements: VRU position, dimensions, velocity and direction. If available, it shall also contain the VRU orientation and its VRU profile. | In case of awareness message for collision risk analysis/avoidance the MCS related exchange enabled via additional data fields metrics (optional) are elaborated in clause 6.9. | [i.1], use case D2. |
| FCOM10 | The VRU ITS-S receiving warning and awareness messages shall check their relevance and if suitable, notify the user appropriately. | The user of a VRU may receive event notifications for neighbouring VRUs. | [i.1], use case B2. |
| NOTE: IVI and MCM, which are more general-purpose messages, have been introduced in ETSI TR 103 300-1 [i.1] and clause 4.3 for similar conditions. SPaT is not considered as a VRU-related warning. | | | |

### 5.2.3 Security requirements

**Table 21: Functional security requirements**

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| FSEC01 | All transmissions to support VRU use cases shall come from an ITS-S and all ITS-S shall ensure that inputs used to form messages are from trustworthy sources. | | |
| FSEC02 | Messages within the VRU system shall use communications security mechanisms appropriate to address the risks that would otherwise be associated with unsecured messages. | | Applies to all use cases [i.1], clause 6; see also security discussion in clause 7.6. |
| FSEC03.1 | The implemented security mechanisms shall be extensible to allow for the deployment of additional use cases. | | Diversity of use cases [i.1], clause 6; see also security discussion in clause 7.6. |
| FSEC03.2 | The implemented security mechanisms shall support the case where vehicular reaction to the VRU is automated, as well as the case where vehicular reaction is caused by alerting the vehicle operator. | | Applies to all use cases [i.1], clause 6; see also security discussion in clause 7.6. |
| FSEC04 | The implemented security mechanisms shall preserve the privacy of all system users to the greatest extent consistent with the safety goals of the system. | | Applies to all use cases [i.1], clause 6; see also security discussion in clause 7.6. |
| FSEC05 | While preserving user privacy (SecFR04), the implemented security mechanisms shall support the ability to detect devices that are sending malicious messages. | | Applies to all use cases [i.1], clause 6; see also security discussion in clause 7.6. |
| FSEC06 | The implemented security mechanisms shall support the ability to withdraw the trusted status of devices that are considered to be causing a risk to the correct operation of the system. | | Applies to all use cases [i.1], clause 6; see also security discussion in clause 7.6. |
| FSEC07 | Special VRU conditions such social weakness, very vulnerable person, disable, pram, wheelchair, etc. shall be security protected against false usage. | | |

## 5.3 Operational requirements

### 5.3.1 System requirements

System operational requirements which characterize the system performances and other operational aspects related to the identified functions.

**Table 22: Operational system requirements**

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| OSYS01 | Positioning capabilities of the communication system shall be better than (50 cm). | The VRUs' relative positioning accuracy will be used to analyse a risk of collision with a vehicle or other VRUs. As VRUs are smaller than vehicles and the trajectory is less predictable, the relative accuracy of VRUs' positions need to be better than for non-VRU vehicles. | [i.1], clause 7.5. |
| OSYS02 | Each VRU and VRU cluster shall have a reference position. | A reference point is necessary to estimate and track the movement of the VRU, combined VRU or VRU cluster. | |
| OSYS03 | The confidence of a VRU dynamic state prediction shall be computed for the purpose of risk analysis. | The prediction of the dynamic state of the VRU is complicated especially for some specific VRU profiles (ex: animal, child, disabled person, etc.). It is thus necessary to associate a confidence level to this prediction as explained in clauses 6.5.10.5, 6.5.10.6 and 6.5.10.9. | |
| OSYS04 | The collision risk analysis requires the information about vertical position. A collision avoidance action should be triggered only in the case where the VRU and vehicle are on the same vertical level (altitude) of the road network. | False positive alert could result of not considering this aspect. An example is when VRUs are on a footbridge crossing a road. | [i.1], use case B2. |
| OSYS05 | The data exchanged shall be recent enough to be useful to the receiver for collision avoidance purposes, leading to a minimum end to end latency time (e.g. less than 300 ms as in the RHS standard ETSI TS 101 539-1 [7]) and to a sufficient data sampling rate (e.g. 10 Hz). | At receiving level (vehicle and VRU) the age of data elements is a key parameter impacting the accuracy of the receiver perception. In particular, at receiving level, a VRU position and velocity are known via their latest received values. But as the VRU device in principle keeps moving, the actual values are changing. Consequently, the data sampling period at the origin and the end to end latency time are key parameters to be minimized. | [i.1], clause 7.5. |
| OSYS06 | A receiver shall take the reported age of data into account when determining the appropriate reaction to a received message. | The correction can be performed via an interpolation mechanism based on a hypothesis of the VRU movement made at this level (e.g. on the mobile trajectory and velocity). | |
| OSYS07 | Battery and other resources (CPU, memory, communication) at the VRU portable device level shall be dimensioned to provide the targeted VRU protection service under knowable conditions. | The resources dimensioning should respond to ITS scalability requirements with the objective to continue offering the VRU basic service when the ITS deployment is progressing. | [i.1], clause 7.5. |
| OSYS08 | If a key resource required for the support of the VRU basic service is defective, whatever the cause, the VRU shall be informed of this failure. | A failure may occur following a lack of energy, a defective component or a cyberattack. | |

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| OSYS09 | The VRU system as defined in clause 3.1 shall be scalable and capable of evolving as necessary to support its deployment without impacting the VRUs' offered services. An already deployed VRU device may be updated or replaced to follow this evolution. | The system needs to be capable to support preventive, curative and evolutive maintenance operations without disturbing the VRU services during all its life cycle. This requires a configuration management capability at the level of all ITS elements. | [i.1], clause 7.7. |
| OSYS10 | The VRU ITS-S shall support communications that are interoperable and as such respect one or several common set(s) of standards. The VRU system interoperability shall be verified before putting VRU system elements on the market. | Interoperability is a key feature of cooperative ITS. | [i.1], clause 6 use cases showing VRU system elements interactions. |
| OSYS11 | The VRU system minimum performances shall be provided and testable before delivery of VRU system elements on the market. | | Minimum performance requirements to be derived from [i.1], clause 7.5 |
| OSYS12 | A device shall be capable of processing an active collision risk event while simultaneously detecting new collision risk events. | This monitoring state may require under some condition the broadcasting of VAMs enabling the collision risk analysis following the reception of the respective current perceptions of involved VRU devices and the derived movement predictions. | [i.1], use case A1, A2, B1, B2, B3, B5. |
| OSYS13 | The VRU movement reliable prediction shall be used to trigger the broadcasting of relevant VAMs when a risk of collision involving a VRU is detected with sufficient confidence to avoid false positive alerts. | See clauses 6.5.10.5, 6.5.10.6 and 6.5.10.9. | [i.1], clauses 4.4, 7.2, 7.5, 7.8 and 7.9. |
| OSYS14 | An ITS station detecting a potential risk of collision with a VRU shall transmit a warning notification message to other road users only if the assessment confidence is higher than 95 %. | 95 % is the value for confidence assessment used in other C-ITS standards for messages such as CAM, DENM or for applications such as the RHS [7]. | [i.1], use case A1. |

### 5.3.2 Communication requirements

**Table 23: Operational communication requirements**

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| OCOM01 | The communication range of a VRU system shall depend on the specific application.<br>Specific values are defined in sub-requirements OCOM01.1 to OCOM01.5 | Here, "communication range" means the distance over which an ITS-S can act in the indicated role based on successful transmission. How to achieve the communication range using the lower layers is out of scope of the present document.<br>The range is defined as 360° round in direct line of sight with at least 95 % probability to receive the packet in 1 s. | [i.1], clause 7.5. |
| OCOM01.1 | A VRU ITS-S that supports communication with an infrastructure ITS-S for VRU protection purposes shall be capable of a transmission range of at least 25 m | This supports the case where an infrastructure ITS-S detects VRU ITS-S by their messages and may transmit the received information in a CPM or even forward the original VAM. | |
| OCOM01.2 | A VRU ITS-S that supports communication with a standard V-ITS-S for pedestrian collision avoidance purposes shall be capable of a transmission range of at least 70 m. | This allows support of VAMs with a TTC of 5 s between a stationary pedestrian and a vehicle moving at 45 km/h. | |
| OCOM01.3 | A VRU ITS-S that supports communication with a standard vehicular ITS-S for cyclist collision avoidance purposes shall be capable of a transmission range of at least 150 m. | This allows support of VAMs with a TTC of 5 s between a cyclist moving at 30 km/h and a vehicle moving at 90 km/h. | |
| OCOM01.4 | A VRU ITS-S that supports communication with a standard vehicular ITS-S for motorcycle collision avoidance purposes shall be capable of a transmission range of at least 300 m. | This is identical to the range used for V2V collision avoidance for standard vehicles. | |
| OCOM01.5 | A VRU ITS-S that supports collision avoidance use cases may be capable of adjusting the transmission range based on local conditions. | To save battery a VRU ITS-S may adjust the transmit power. For example, a reasonable transmit power algorithm would base the anticipated transmit range on a TTC of 5 s with the highest-speed vehicle detected in the previous 5 s (possibly taking maps into account so that, for example, vehicles on a freeway on the other side of a wall do not cause the device to use more transmit power than would be useful). | |
| OCOM02 | Communication latency shall be less than 5 ms under open sky conditions with unobstructed transmission from sender to receiver | This is the latency between the start of the request to transmit a packet of data at access layer level until reception at access layer level in the peer. The requirement specifies "open sky conditions" and "unobstructed transmission" to make clear the test conditions related to this requirement; it will of course be the case that any implementation that meets the requirements under these conditions will also meet the requirements under a wide range of less favourable conditions. | [i.1], clause 7.5. |
| OCOM04 | The VAM shall support dynamic length. | The length of the message depends on the content of its data elements and the actually deployed data elements<br>This is according to standard messages requirements and takes consideration of the security needs. This is detailed in ETSI TS 103 300-3 [i.11]. | |

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| OCOM05 | The VAM shall support dynamic timing. | The timing of VAM transmission depends on the context of the VRU.<br>This is according to standard messages requirements and takes consideration of the security needs. This is detailed in ETSI TS 103 300-3 [i.11]. | |

### 5.3.3 Security requirements

**Table 24: Operational security requirements**

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| OSEC01 | The security processes shall support generating messages at a rate of 10 Hz, receiving messages at a rate of 2 KHz, latency of 300 ms end-to-end and an average sent packet size over 1 s of 300 bytes. | | Applies to all use cases [i.1], clause 6; see also security discussion in clause 7.6. |

## 5.4 Additional recommendations

**Table 25: Additional recommendations**

| Rec-Id | Recommendation text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| REC01 | A VRU ITS-S should be able to perform collision risk assessment if sufficient battery and processing power are available in the VRU device. | The VRU may not have sufficient processing power or battery to make this assessment. Accordingly, this is not a requirement for the VRU device. | Use case A2. |
| REC02 | All elements of the VRU system should comply with the relevant spectrum regulation and related standards. | Interoperability between interacting VRU system elements requires standardization of communication profiles common to all participating elements. | |
| REC03 | A VRU device should be capable of processing at least 1 000 incoming messages per second without degradation of its ability to detect potential collision risks. | Same as in the RHS minimum performance requirements [7] | |

# 6 Functional architecture of the VRU system

## 6.1 VRU profile specification

In this clause four basic VRU profiles are specified. These profiles are the basis for the further definition of the VRU functional architecture. The profiles are derived from the use cases and from the analysis in clause 7.2 of ETSI TR 103 300-1 [i.1].

A VRU can only be a living being. This living being is considered as a VRU only when it is in the context of a safety related traffic environment.

So, a living being in a house is not a VRU but as soon as it comes closer to the street (e.g. 2 m or 3 m), it is part of the safety related context and is in a role of a VRU.

This consideration is important since it will limit the amount of communications. A C-ITS communications device needs only to start acting as a VRU ITS-S when the living being associated with it starts acting in the role of a VRU.

**Profile classification parameters.**

- Maximum and average (typical) speed values (it may be associated with its standard deviation).
- Minimum and average (typical) communication range: The communication range is calculated based on the assumption that an awareness time of 5 s is needed to warn/act on the traffic participants.
- Environment e.g. type of area (urban, sub-urban, rural, highway).
- Average Weight and standard deviation.
- directivity/trajectory ambiguity**:** it gives the level of confidence in the predictability of the behaviour of the VRU in its movements.
- Cluster size: Number of VRUs in the cluster. A VRU may be leading a cluster and then indicate the cluster size (see clause 6.2.1). The cluster reference position is defined as described in clause 6.2.3.

NOTE 1: These parameters are not dynamic parameters maintained in internal tables, but indications of typical values used to classify the VRUs into behaviour profiles and evaluate the behaviour of a VRU belonging to a specific profile.

NOTE 2: Version V1.1.1 of ETSI TS 103 300-3 [i.11] recommends VRU clustering operations when transmitting the VAM for pedestrians only. Roadside equipment can also report VRU clusters through the CPM.

The values given in the VRU profile tables below are only indicative.

**VRU Profile 1 - Pedestrian.**

Typical VRUs in this profile: pedestrians, i.e. road users not using a mechanical device for their trip. It includes for example pedestrians on a sidewalk, but also children, prams, disabled persons, blind persons guided by a dog, elderly persons, riders off their bicycles, road workers, first responders.

**Table 26: Pedestrian VRU profile**

| Parameter | Values | Typical value in operation | Comment |
|---|---|---|---|
| Speed | < 12 km/h<br>< 25 km/h | 5 km/h | This profile also includes running pedestrians and animals. |
| Communication range | Minimum supported maximum value: 70 m | 50 m urban and sub-urban<br>250 m rural (if capable) | Main usage in urban and sub-urban environment lead to the typical range of 50 m. |
| Environment | Urban and sub-urban<br>Rural<br>Highway | Urban/sub-urban | Mainly in urban and sub-urban environment<br>Highway after accident/ Road works. |
| Weight class | Low: < 10 kg<br>Medium: < 30 kg<br>High: > 30 kg | High (80 kg) | The weight class is intended to distinguish between small children, larger children and adults. |
| Trajectory ambiguity | High | High | |
| Cluster size | Up to 50 | > 1 | 50 is an arbitrary number brought in by common sense and performance requirements in clause 5. Above 50, the management of the cluster would be challenging.<br>This may be changed under further study (see ETSI TS 103 300-3 [i.11], clause 8). |

**VRU Profile 2 - Bicyclist.**

Typical VRUs in this profile: bicyclist and similar e.g. light vehicles riders, possibly with an electric engine. It includes bicyclists, but also wheelchair users, horses carrying a rider, roller-skaters, e-scooters, personal transporter, pedelec and speed-pedelec riders, etc.

NOTE 3: It has to be noted that a light vehicle itself does not represent a VRU but only the combination with a person will create the VRU.

**Table 27: Bicyclist VRU profile**

| Parameter | Values | Typical value in operation | Comment |
|---|---|---|---|
| Speed | < 25 km/h (all environments) | 20 km/h | Higher typical speeds do not have an impact on the profile. |
| Communication range | Minimum supported maximum value: 150 m | 70 m urban and sub-urban<br>250 m rural | Main usage in urban and sub-urban environment lead to the typical range of 50 m.<br>Higher typical speed leads to higher range. |
| Environment | Urban and sub-urban<br>Rural | Urban/Sub-urban | Mainly in urban and sub-urban environment. |
| Weight class | Low: < 15 kg<br>Medium: < 40 kg<br>High: > 40 kg | High (90 kg) | The weight class is intended to distinguish between small children, larger children and adults.<br>Weight includes the driver (bicycle alone is not a VRU). |
| Trajectory ambiguity | Medium to high | Medium | |
| Cluster size | 20 | > 1 | 20 is an arbitrary number brought in by common sense and performance requirements in clause 5. Above 20, the management of the cluster would be challenging.<br>This may be changed under further study (see ETSI TS 103 300-3 [i.11], clause 8). |

**VRU Profile 3 - Motorcyclist.**

Typical VRUs in this profile: motorcyclists, which are equipped with engines that allow them to move on the road. It includes users (driver and passengers, e.g. children and animals) of Powered Two Wheelers (PTW) such as mopeds (motor scooters), motorcycles or sidecars.

NOTE 4: It has to be noted that a PTW itself does not represent a VRU but only the combination with a person will create the VRU.

**Table 28: Motorcyclist VRU profile**

| Parameter | Values | Typical value in operation | Comment |
|---|---|---|---|
| Speed | < 45 km/h (mopeds)<br>Same speed as passenger cars (motorcycles) | 35 km/h (mopeds)<br>Same speed as passenger cars (motorcycles) | 45 km/h is the maximum speed of a scooter. |
| Communication range | Minimum supported maximum value: 300 m | 70 m urban and sub-urban<br>250 m rural | A motorcycle sends CAMs and is subject to the performance requirements related to CAMs. |
| Environment | Urban and sub-urban<br>Rural<br>Highway | Urban/Sub-urban | Mainly in urban and sub-urban environment. |
| Weight class | Low: < 25 kg<br>Medium: < 140 kg<br>High < 300 kg | Medium (140 kg) | The weight class is intended to distinguish between different types of PTWs. Weight includes the driver (motorcycle alone is not a VRU). |
| Trajectory ambiguity | Low to medium | Low | |
| Cluster size | 20 | > 1 | 20 is an arbitrary number brought in by common sense and performance requirements in clause 5. Above 20, the management of the cluster would be challenging.<br>This may be changed under further study (see ETSI TS 103 300-3 [i.11], clause 8). |

**VRU Profile 4 - Animals presenting a safety risk to other road users.**

Typical VRUs in this profile: dogs, wild animals, horses, farm animals: cows, sheep, service animals, etc. Some of these VRUs might have their own ITS-S (e.g. service dog in a city or a horse) but most of the VRUs in this profile will only be indirectly detected and reported with messages such as the CPM. Especially wild animals in rural areas and highway situations.

Clusters of animals VRU might be herds of animals, like a herd of sheep or cows or wild boars. This profile has a lower priority when decisions have to be taken to protect a VRU.

**Table 29: Animal VRU profile**

| Parameter | Values | Typical value | Comment |
|---|---|---|---|
| Speed | < 40 km/h (all environments) | 5 km/h | |
| Communication range | Minimum supported maximum value: 70 m. | 70 m urban and sub-urban<br>250 m rural | |
| Environment | Urban and sub-urban<br>Rural | Urban/sub-urban/rural | In urban and sub-urban environment, applies mainly to individual animals, e.g. dogs. In Rural environments, applies to individual animals (e.g. horses) but also to clusters of animals (e.g. cows, sheep). |
| Weight class | Low: < 15 kg<br>Medium: < 40 kg<br>High: > 40 kg | Medium (25 kg) | The weight class is intended to distinguish between small animals like a cat, medium animals like a dog or a big animal like a wild boar or a horse. |
| Trajectory ambiguity | high | High | |
| Cluster size | Up to 100 | > 1 | See ETSI TS 103 300-3 [i.11], clause 8. |
| Priority | Below profiles 1 to 3 | | |

# 6.2 VRU cluster definition

## 6.2.1 VRU cluster concept

NOTE: The present clause defines the notion of VRU cluster. Its detailed specification is part of the VRU basic service standard (ETSI TS 103 300-3 [i.11]).

The number of VRUs operating in a given area can get very high (up to 500, see UC-B2, VRU crossing a road, and clause 7.4.1 in ETSI TR 103 300-1 [i.1]) or the VRU can be combined with a VRU vehicle (e.g. rider on a bicycle). In order to reduce the amount of communication and thus the resource usage, VRUs should be grouped together leading to VRU clusters. These clusters can be homogeneous VRU clusters (group of pedestrians) or heterogeneous VRU clusters (groups of pedestrians and bicyclists). These clusters are considered as a single object/entity and only the cluster leader will continuously transmit the VAM. The parameters of the VRU cluster will be communicated using the VAM. The VAM message contains an optional container that indicates whether the VRU is leading a cluster, which is not present for an individual VRU (other VRUs in the cluster should not transmit VAM). The leading VRU also indicates in the VAM whether it is a homogeneous cluster or heterogeneous, the latter one being of any combination of VRU profiles.

Indicating heterogeneous clusters is important since it provides useful information about trajectory and behaviours prediction when the cluster is broken up.

The use of a bicycle or motorcycle will significantly change the behaviour and parameters set of the VRU using this non-VRU object (or VRU vehicle) "bicycle"/"motorcycle". A combination of a VRU and a non-VRU object is called combined VRU. A combined VRU can be created only in the case where both VRU and VRU vehicle are equipped with VRU-St devices.

The goal of a VRU cluster is to reduce the communication requirements (e.g. spectrum requirements). In the case of the combined VRU, the change of the VRU role is also an important goal.

A VRU cluster is a set of 2 or more VRUs (e.g. pedestrians) such that the VRUs move in a coherent manner, i.e. with coherent velocity or direction and within a bounding box. VRUs with VRU Profile 3 (motorcyclists) are not involved in the VRU clustering concept.

**Coherent "cluster" velocity:** velocity range of VRUs in a cluster such that the differences in speed and heading between any of the VRUs are below a predefined threshold (see ETSI TS 103 300-3 [i.11]).

**VRU bounding box:** a geometric shape containing all the VRUs in the cluster and such that all the VRUs in the bounding box make contact with the surface at approximately the same elevation.

## 6.2.2 VRU cluster requirements

**Table 30: VRU cluster requirements**

| Req-Id | Requirement text | Explanation | UC reference from ETSI TR 103 300-1 [i.1] |
|---|---|---|---|
| RCL01 | A VRU device (with VRU profile 1, 2 or 4) shall be able to determine whether it should be acting as a clustered or individual VRU. | See also RCL02. | |
| RCL02 | Only the VRU-St device type (see clause 4.1) shall be able to create and lead a cluster. The VRU device type (St or TX) shall be indicated in the VAM. | As VRU-TX only devices cannot receive and decode the VAMs of other VRU ITS-S, they are not able to participate in a cluster. | |
| RCL03 | A VRU-St shall continuously detect whether or not it is part of a VRU cluster. | | |
| RCL04 | If the VRU is part of a VRU cluster, only the VRU leading the cluster shall transmit the VAM(s). | | Use case B1. |
| RCL05 | The VRU cluster leader shall transmit the VAM(s), indicating the cluster total dimension and velocity. | | |
| RCL06 | If the VRU is part of a VRU cluster and not leading the cluster, the VRU shall transmit a VAM indicating that it is joining the cluster and stop the transmission of VAM. | | |
| RCL07 | If the VRU is no longer part of the VRU cluster which it belonged to, it shall resume the transmission of the VAM(s). | | Use case B3. |
| RCL08 | If the VRU is part of a cluster, it shall continuously monitor VAM from the cluster leader. If there is no VAM from cluster leader for a pre-defined amount of time, VRU assumes the cluster leader to be lost and the VRU shall resume sending VAM. | | |

## 6.2.3 VRU cluster reference position

A VRU cluster is considered as a single entity in the overall VRU communication. Only a single cluster VAM is generated per cluster containing the relevant information/parameter of the cluster. These parameters are (more details can be found in ETSI TS 103 300-3 [i.11]):

- Cluster ID (in case several clusters are collocated).
- Number of participants in the cluster, $N_{\text{cluster}}$.
- Shape and dimension of the cluster bounding box.
- Speed of cluster (identical to the speed of the clusre leader).
- Position of cluster (position of the cluster leader, the VRU cluster reference position).

VRU reference position: The reference position of a cluster of VRUs shall refer to ground position at the centre point of the face side of the cluster bounding box. This is consistent with the definition of reference points in ETSI EN 302 890-2 [5] (PoTi) and may be different from the position of the cluster leader, as VRUs may move inside the cluster.

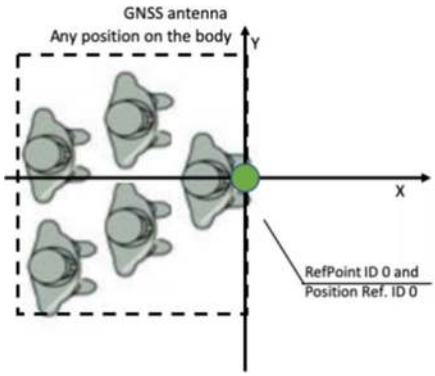



**Figure 3: Example reference position in the case of a rectangle cluster of pedestrians**

### 6.2.4 VRU cluster approach

Clustering of VRUs is one of the possible solutions for prevention and reduction of the number of transmitted messages (see requirement FCOM03 for all possible solutions identified).

A VRU device is able to determine whether it should be acting as a clustered or individual VRU (see note 2 below). If a VRU is part of a cluster, only the VRU leading the cluster, i.e. cluster leader VRU, transmits the VAM, indicating the cluster's total dimension and velocity. A cluster leader VRU leads and manages a cluster, e.g. creating and breaking up a cluster, and cluster member VRUs join and leave a cluster.

NOTE 1: The VRU cluster operation is fully specified in ETSI TS 103 300-3 [i.11].

The following operations have been identified in the present document:

- **Creating a VRU cluster:** When a VRU determines to create a cluster based on the received VAMs from other VRUs, it sends a VAM indicating that it will lead the cluster with the VRU cluster's identifier.
- **Breaking up a VRU cluster:** When a VRU determines to break up a cluster, it sends a VAM indicating that it will break up the cluster with the VRU cluster's identifier. The cluster leader sends breaking up indication a pre-defined amount of time in consecutive VAMs.
- **Joining a VRU cluster:** When a VRU receives VAMs from a cluster leader VRU, the VRU analyses the received VAMs and decides whether it should join the cluster or not. In order to join the cluster, the VRU sends a VAM indicating that it is joining the cluster with the VRU cluster's identifier (repeated a pre-defined amount of time), stops transmission afterwards, and monitors the VAMs from the cluster leader VRU. This allows the cluster leader VRU to know whether the cluster is homogeneous or heterogeneous, and the profile, size, speed and reference position of the cluster. If the new VRU member finds that cluster leader has not heard its last VAM with cluster join indication (e.g. based on bounding box information transmitted in next VAM from the cluster leader), it should send VAM again with cluster leave indication and resume its individual VAM transmission.
- **Leaving a VRU cluster:** When a VRU in a cluster receives VAMs from the cluster leader VRU, the VRU analyses the received VAMs and decides whether it should leave the cluster or not (see note 3 below). In order to leave a cluster, it sends a VAM, i.e. first VAM after its silence, indicating that it is leaving the cluster and resumes the transmission of the VAM. A VRU is always allowed to leave a cluster for any reason.
- **Determining cluster leader lost:** In some cases, the cluster leader may lose communication connection or fail as a node. In this case, the cluster leader cannot send VAM any more on behalf of the cluster. If there is no VAM for a pre-defined amount of time, VRU(s) assume(s) the cluster leader to be lost and the VRU(s) leave the cluster as described previously.

NOTE 2: Only the VRU-St device type (see clause 4.1) is allowed to be part or lead a cluster. The VRU device type (VRU-St or VRU-TX) is indicated in the VAM.

NOTE 3: A VRU device determines whether it can join or should leave a cluster by comparing its measured position and kinematic state with the position and kinematic state indicated in the VAM of the cluster leader VRU. If the compared information fulfils certain conditions, e.g. less than 3 - 5 meters away and speed difference less than 5 % of own speed, the VRU device can join the cluster. After joining the cluster, when the compared information does not fulfil the conditions any longer, the VRU device leaves the cluster. In some cases, moving VRU clusters on sidewalk with similar coherent cluster velocity profiles may come closer with fully or partially overlapped bounding boxes. In such a case, merging these VRU clusters can further reduce VRU messaging in the network. Merging clusters may be performed simply by breaking up one of the clusters, which enable VRUs to join the other cluster.

## 6.3 VRU system functional architecture

The present document specifies the VRU-related functional entities residing in the facilities and application layers of an ITS Station, as defined in ETSI EN 302 665 [1] (ITS station reference architecture) and ETSI TS 102 894-1 [2] (Facilities layer specification) that are relevant for the operation of the VRU system.

Figure 4 provides the global architecture of the VRU system, based on four levels, one for each category of ITS-S (personal, vehicle, roadside, central). When present, each level of the VRU system shall include an ITS-S for that level in addition to other relevant components. Functional entities in Figure 4 located beside the ITS-S are not directly related to C-ITS (e.g. the HMI). Devices not including an ITS-S are considered only according to their relationship with an ITS-S. More details are provided in clause 6.5.

NOTE 1: This architecture is inspired from the architecture described by the VRUITS project [i.2].

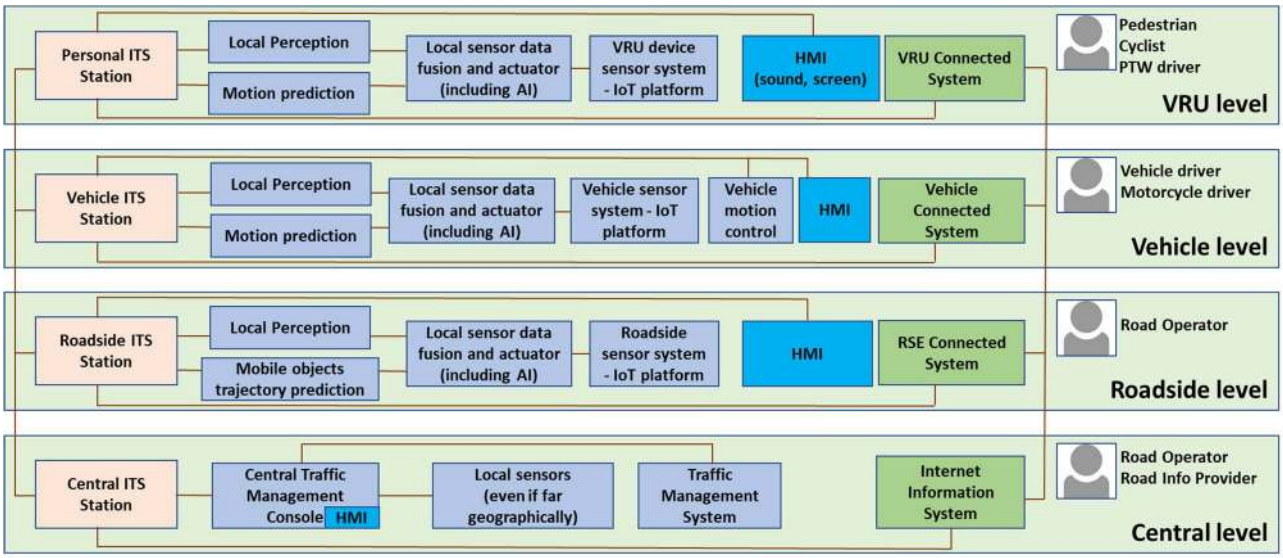



**Figure 4: VRU system functional architecture**

For each level, Figure 4 shows from right to left the entities which operate at the same level but are NOT included in the ITS-S:

- The relevant users at that level.
- The relevant HMI.
- Vehicle motion control (at vehicle level only) for automated vehicles. This entity is optional and does not appear in human driven vehicles. Both HMI and vehicle motion control entity may be triggered by the ITS-S applications.
- Local device sensor system and IoT Platform that collects and shares IoT data. The sensor system can be composed of one or more cameras, radars, lidars, etc. in a vehicle or roadside equipment. In the central station, it consists of sensors that may be located on the side of the road, but that directly report their data to the central station, without the involvement of a V-ITS-S or a R-ITS-S. In the VRU device, it consists mainly of the gyroscope and accelerometer.

- Local device sensor fusion and actuator application, which may contain Artificial Intelligence (AI) and aggregates the data flow issued by the sensor system.
- Local perception and trajectory prediction applications which consume the output of the fusion application and feed the ITS-S applications.
- The relevant ITS-S, which contains all or part of the entities shown in Figure 5.

NOTE 2: Local perception is obtained from the output of the local sensors data fusion and AI applications entities.

## 6.4 VRU-related functions in the ITS station architecture

Figure 5 zooms in the ITS-S element of Figure 4 and shows the main components of the ITS-S architecture specified in ETSI EN 302 665 [1] that directly impact the VRU system operation. It introduces a new functional entity described in more details in ETSI TS 103 300-3 [i.11], the VRU Basic Service. Other functions as defined in ETSI TS 102 894-1 [2] may be present and support the operation of these services, e.g. the communication support facilities or the HMI support. Figure 5 also illustrates a possible functional split at the application layer level of the ITS-S. More details are provided in clause 6.5.

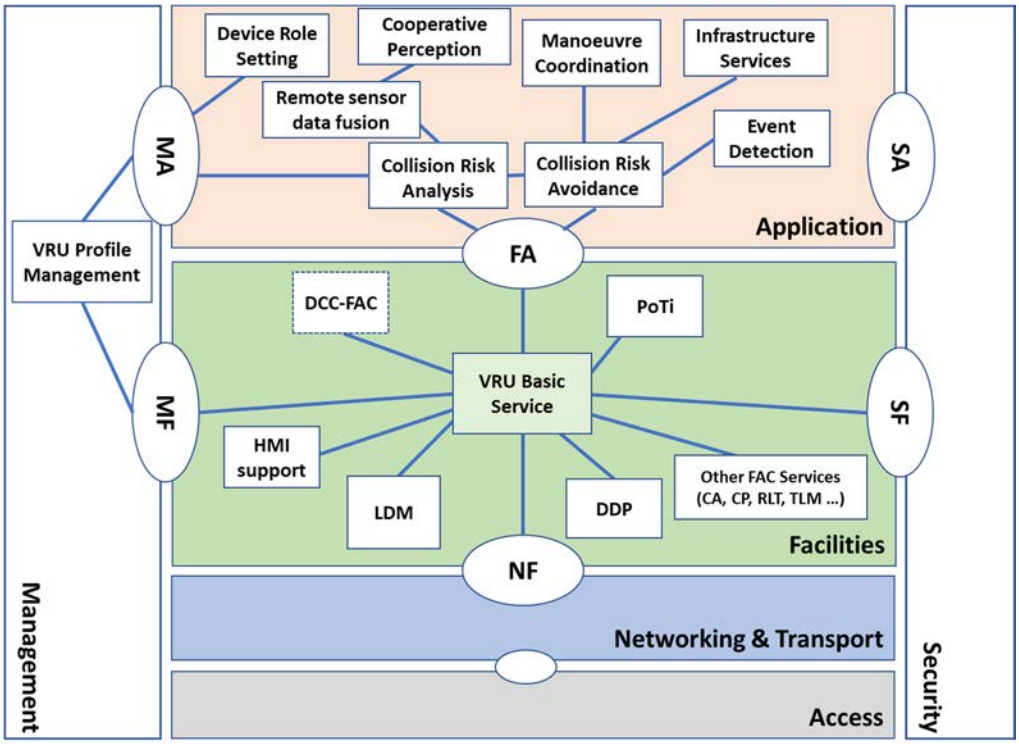



**Figure 5: VRU functionality mapped to the ITS Station architecture**

The VRU-specific functionality, including interfaces mapped to the ITS-S architecture, is shown in Figure 5. It shall be centred around the **VRU Basic Service** located in the facilities layer; it shall consume data from other services located in the facilities layer such as PoTi and LDM, Data Provider (DDP) and shall be linked with other entities such as application support facilities, e.g. CA service, Collective Perception Service, Infrastructure service, etc. The VRU basic service shall be responsible to transmit the VAM message, to identify whether the VRU is part of a cluster and to enable the assessment of a potential risk of collision by ITS applications.

The VRU Basic Service shall interact with the VRU profile management entity in the management layer to learn whether the ITS-S has the VRU role enabled (device user is considered as a VRU) or switched OFF.

The application layer shown in Figure 5 recommends a possible distribution of functional entities that would be involved in the protection of VRUs, based on the analysis of VRU use cases.

## 6.5 Function descriptions

### 6.5.1 General overview

Clause 6.5 describes the functions illustrated in Figure 4 and Figure 5.

The proposed functional descriptions are conceptual models which can be implemented in different ways. The specification of exposed interfaces would contribute to make the ITS functionally testable and measurable in terms of minimum performances requirements.

This clause proposes backward compatible evolutions of the communication architecture defined in ETSI EN 302 665 [1] to support the VRU basic service, mainly focusing on the application layer, facilities layer and the management plane (see Figure 5).

### 6.5.2 Sensor system - IoT platform

The VRU sensor system is at least composed of the PoTi data provider present in each ITS-S of the system. The PoTi entity provides the global time common to all VRU system elements and the real time position of the mobile elements of the VRU system.

Local sensors may also be embedded in the mobile elements as well as in the road infrastructure (e.g. camera in a smart traffic light). The IoT platform which can be distributed over the system elements may contribute to provide additional information related to the environment surrounding the VRU system and its context.

### 6.5.3 Local Sensor data fusion and actuator (including AI)

The sensor data fusion function is the fusion of local perception data obtained from different local sensors. Sensor data fusion relies on the consistency of its inputs and on their timestamping which needs to correspond to one common given time.

### 6.5.4 Local perception

The local perception function is provided by the local processing of information collected by local sensor(s) associated to the system element (VRU device, vehicle, RSE, central systems) considered. Its main purpose is to detect and characterize objects (static and mobile) which are likely to cross the trajectory of the considered moving objects.

The infrastructure and particularly the road infrastructure may offer services relevant to the VRU support service. The infrastructure may have its own sensors detecting VRUs evolutions and then computing a risk of collision if also detecting local vehicles' evolutions, either directly via its own sensors or remotely via a cooperative perception supporting services such as the CPS (ETSI TR 103 562 [i.4]).

Generally, road marking (e.g. zebra crossings) and vertical signs need to be considered to increase the confidence level associated with the VRU detection and mobility as normally VRU have to respect these marking/signs.

### 6.5.5 Motion dynamic prediction

The motion dynamic prediction function is related to the behaviour prediction of the considered moving objects. Motion dynamic includes the moving object trajectory resulting from evolution of the successive mobile positions. A change of the moving object trajectory or of the moving object velocity (acceleration/deceleration) impacts the motion dynamic prediction. In most cases, when VRUs are moving, they still have a large amount of possible motion dynamic in terms of possible trajectories and velocities. So, the problem of motion dynamic prediction is to identify as quickly as possible which motion dynamic will be selected by the VRU and if this selected motion dynamic is subject to a risk of collision with another VRU or a vehicle.

The motion dynamic prediction function analyses the evolution of mobile objects which trajectories may meet at a given time and so create a risk of collision between them.

The motion dynamic prediction works on the output of cooperative perception considering:

- The current trajectories of considered VRUs for the computation of the path prediction.

- The current velocities and their past evolutions for the considered mobiles for the computation of the velocity evolution prediction.

- The reliability level which can be associated to these variables.

In many cases, working only on the output of the cooperative perception is not sufficient to make a reliable prediction because of the uncertainty which exists in terms of VRU trajectory selection and its velocity. However, complementary functions may assist in increasing consistently the reliability of the prediction, for example:

- The use of the VRU device navigation system which provides assistance to the VRU to select the best trajectory for reaching its planned destination. With the development of MaaS (Mobility as a Service, see note below), multimodal itinerary computation may also indicate to the VRU dangerous areas and then assist the motion dynamic prediction at the level of the multimodal itinerary provided by the system.

- The knowledge of the VRU habits and behaviours. Very often VRUs follow the same itineraries, using similar motion dynamics, for example when going to the main POI (Point of Interest) which is related to their main activities (e.g. going to school, going to work, doing some shopping, going to the nearest public transport station from their home, going to sport centre, etc.). The VRU device or a remote service centre may learn and memorize these habits.

- The indication by the VRU itself of its selected trajectory in particular when changing it (e.g. using a right turn or left turn signal similar to vehicles when indicating a change of direction).

NOTE: MaaS is a new European developing concept supporting the development of transport multi-modality. This service includes the calculation of multimodal transport itineraries proposed to users for going from one origin to a final destination. In the case of a VRU, a multimodal transport itinerary is composed of a succession of multimode transport segments, some of them involving VRU motions (walking or using a bicycle/scooter). These VRU motion segments can be the object of clearly described itineraries stressing the collision risk areas located on the VRU path.

The output of this function is provided to the risk analysis function.

### 6.5.6 Vehicle motion control

The vehicle motion control is a function under the responsibility of the human driver or of the vehicle if it is able to drive in automated mode.

It is capable to act on the trajectory of the vehicle or on its speed.

### 6.5.7 Human - Machine Interface (HMI)

The Human Machine Interface (HMI) is not mandatory, but, when present, it enables:

- The configuration of initial data (parameters) in the management entities (e.g. VRU profile management) and in other functions (e.g. VRU basic service management).

- The communication to the device owner of external events related to the VRU basic service.

- Signalling a risk of collision (TTC > 2 s) detected by at least one element of the system.

- Alerting about an immediate risk of collision (TTC < 2 s) detected by at least one element of the system.

NOTE: The Time To Collision (TTC) value evaluation and how it can be used is described in more details in clause 6.5.10.5.

In the case of a VRU, similar to a vehicle driver, the HMI provides the information to the VRU, considering its profile (e.g. for a blind person, the information is presented with a clear sound level). In most of the cases, it is recommended to use visual and audible information to VRUs, according to the capabilities of the VRU devices.

### 6.5.8 Connected System/Information System

The VRU system is a connected system made of up to 4 different levels of equipment as represented in Figure 4. It is also an information system which collects in real time information resulting from events, processes the collected information and stores them together with processed results.

At each level of the VRU system, the information collection, processing and storage is related to the functional and data distribution scenario which is implemented.

### 6.5.9 Traffic Management System

Traffic Management System (TMS) collects data from interconnected system elements, then processes them and provides traffic information to road users. The processing of collected data is performed for well-defined services and supporting applications.

In the case of VRU protection, a TMS allows to collect data and disseminate the presence of VRUs in a larger area than a single roadside equipment. This may be useful for example in the case of a highway where vehicles are driving at high speed and need to be informed of a potential risk of collision earlier than on a rural road, or in smart cities where monitoring specific areas allows to inform drivers that a march is ongoing on a specific street.

In case of the MaaS deployment, the Traffic Management System can be transformed/extended into a Mobility Service management centre providing multimodal transport navigation and associated payment platform. Multimodal transport navigation also considers the space/time associated to road users' vulnerability (e.g. when pedestrian or using a soft transportation mean such as bicycle, scooter, etc.

### 6.5.10 ITS Station Application Layer

#### 6.5.10.1 Global overview

The functions described below are proposed to be located at the application layer level. However, if there is a generic and justified need to share them between several applications, part of these functions may be moved to the facilities layer.

#### 6.5.10.2 Device role setting

A VRU can be equipped with a portable device which needs to be initially configured and may evolve during its operation following context changes which need to be specified. This is particularly true for the setting-up of the VRU profile and sub-profile which can be achieved automatically at power on or via an HMI. The change of the road user vulnerability state needs to be also provided either to activate the VRU basic service when the road user becomes vulnerable or to de-activate it when entering a protected area. The VRU basic service remains operational whether the user's role as a VRU is enabled or not.

The initial configuration can be set-up automatically when the device is powered up, for example for the VRU device type which may be:

- VRU-Tx with the only communication capability to broadcast messages (yet complying with the channel congestion control rules).
- VRU-Rx with the only communication capability to receive messages.
- VRU-St with full duplex communication capabilities.

During service operation, the VRU profile may also change due to clustering or cluster breaking up. Accordingly, the VRU device role will be able to evolve according to the VRU profile changes.

NOTE: The device role setting specification is out of scope of the present document and should be covered in another specification to enable to operation of VRU devices.

#### 6.5.10.3 Remote sensor data fusion and actuator (including AI)

The local perception obtained by the computation of data collected by local sensors may be augmented by remote data collected by elements of the VRU system (vehicles, RSE) via the ITS-S. These remote data are transferred using standard services such as the CPS. In such case it is necessary to fuse these data. The data fusion may provide at least three possible results:

- After a data consistency check, the received remote data are not coherent with the local data. In this case, the system element has to decide which source of data can be trusted and ignore the other.
- Only one input is available (e.g. the remote data). This means that the other source does not have the possibility to provide information. In this case, the system element may trust the only available source.
- After a data consistency check, the two sources provide coherent data which augment the individual inputs provided.

The use of AI is necessary to recognize and classify the detected objects (VRU, motorcycle, type of vehicle, etc.) but also their associated dynamics. The AI can be located in any element of the VRU system.

The same approach is applicable to actuators, but in this case, the actuators are the destination of the data fusion.

#### 6.5.10.4 Cooperative Perception

As shown in Figure 6, the perception chain can be the fusion of the results of several perception functions at predefined times:

- The local perception is provided by the collection of information from the environment of the considered ITS element (VRU device, vehicle, infrastructure). This information collection is performed using relevant sensors (optical camera, thermal camera, radar, lidar, etc.).
- The remote perception is provided by the provision of perception data via C-ITS (mainly V2X communication). Basic services such as the Cooperative Awareness or the Collective Perception Service can be used to transfer a remote perception.
- Several perception sources may be used to achieve the cooperative perception function. It is then necessary to verify the consistency of these sources at predefined instants and if not consistent, select the best one according to the confidence level associated with each perception variable.
- The result of the cooperative perception should comply with the required level of position accuracy as specified by PoTi in ETSI EN 302 890-2 [5].

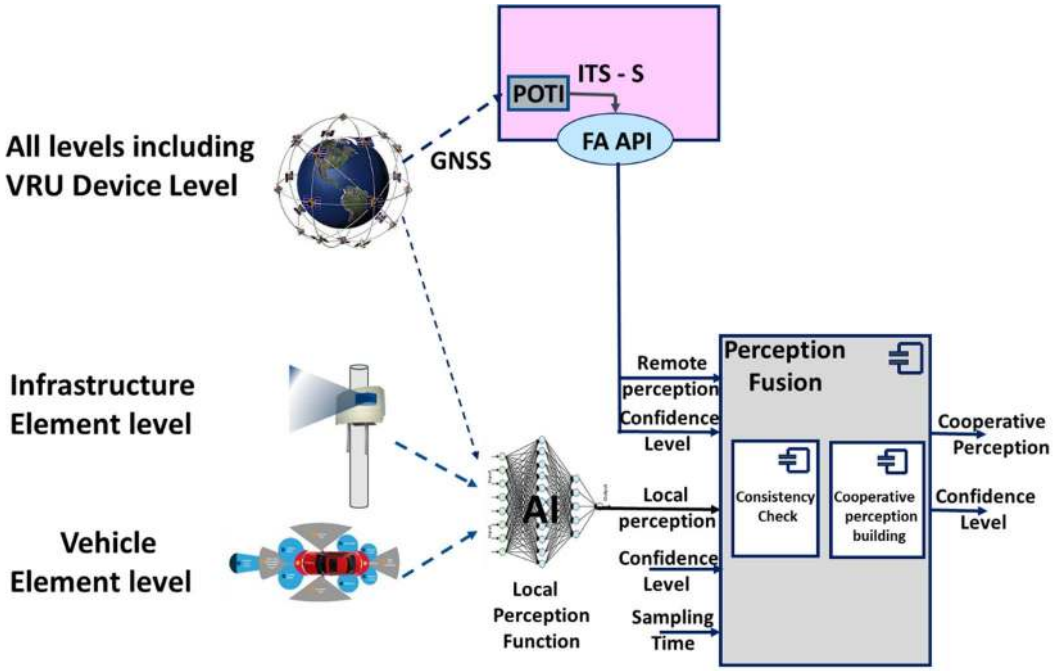



**Figure 6: Example of a cooperative perception functional model**

An associated confidence level is necessary to build the cooperative perception resulting from the fusion in case of differences between the local perception and the remote perception. It will be also necessary for the exploitation by other functions (e.g. risk analysis) of the cooperative perception result.

The perception functions from the device local sensors processing to the end result at the cooperative perception level may present a significant latency time of several hundred milliseconds. For the characterization of a VRU trajectory and its velocity evolution, there is a need for a certain number of vehicle position and velocity measurements thus increasing the overall latency time of the perception. Therefore, it is necessary to estimate the overall latency time of this function to take it into account when selecting a collision avoidance strategy.

### 6.5.10.5 Collision risk analysis

The collision risk analysis function analyses the motion dynamic prediction of the considered moving objects associated to their respective levels of confidence (reliability). The objective is to estimate the likelihood of a collision and then to identify as precisely as possible the Time To Collision (TTC) if the resulting likelihood is high. Other variables may be used to compute this estimation, as described below. Figure 7 provides an example of TTC calculation and use.

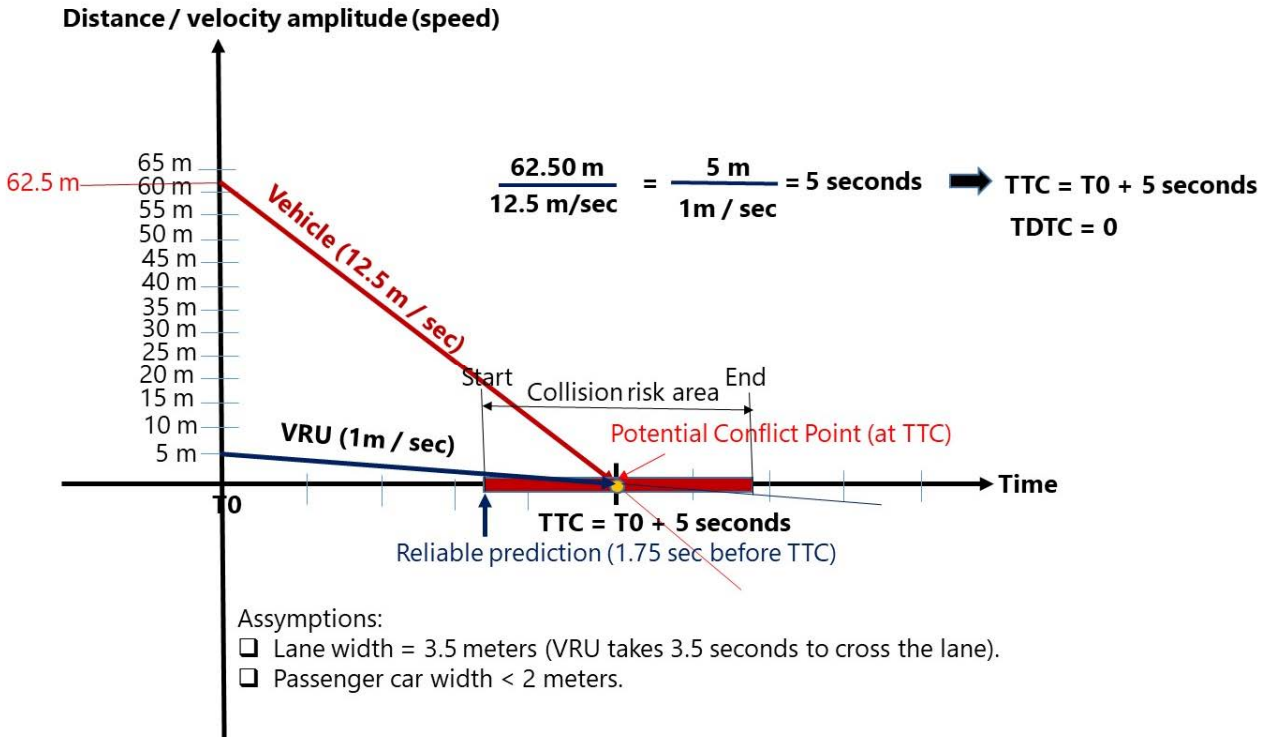



**Figure 7: Example of TTC calculation**

The following two conditions are mandatory to be able to calculate the TTC:

- Two or more considered moving objects follow trajectories which intersect somewhere at a position which can be called "potential conflict point".
- If the moving objects maintain their motion dynamics (trajectories, speeds) it is possible to predict that they will collide at a given time which can be estimated through the computation of the time (called TTC) necessary for them to arrive simultaneously at the level of the identified potential conflict point.

From Figure 7, it can be derived that:

- A TTC prediction can only be reliably established when the VRU enters the collision risk area, that is to say very late in this example (1,75 s before reaching the potential conflict point). This is due to the uncertainty nature of the VRU pedestrian motion dynamic (mainly its trajectory) before deciding to cross the road.
- At the potential conflict point level, another measurement, the TDTC (Time Difference for pedestrian and vehicle travelling to the potential conflict point) can be used to estimate the collision risk level. For example, if it is not acted on the motion dynamic of the pedestrian or/and on the motion dynamic of the vehicle, TDTC is equal to ZERO and the collision is certain. Increasing the TDTC reduces the risk of collision between the VRU and the vehicle.
- The potential conflict point is in the middle of the collision risk area which can be defined according to the lane width (e.g. 3,5 m) and vehicle width (maximum 2 m for passenger cars).

The TTC is one of the variables that can be used to define a collision avoidance strategy and the operational collision avoidance actions to be undertaken. Other variables may be considered such as the road state, the weather conditions, the triple of {Longitudinal Distance (LoD), Lateral Distance (LaD), Vertical Distance (VD)} along with the corresponding threshold triple of {MSLaD, MSLoD, MSVD}, TII (explained in clause 6.5.10.9) and the capabilities of the mobile objects to react to a collision risk and avoid a collision.

NOTE: The collision risk analysis function compares LaD, LoD and VD, with their respective predefined thresholds, MSLaD, MSLoD, MSVD, respectively. If all the three metrics are simultaneously less than their respective thresholds, that is LaD < MSLaD, LoD < MSLoD, VD < MSVD, then the collision avoidance actions would be initiated. Those thresholds could be set and updated periodically or dynamically depending on the speed, acceleration, type, and loading of the vehicles and VRUs, environment and weather conditions. On the other hand, the TII reflects how likely is the ego-VRU ITS-S trajectory going to be intercepted by the neighbouring ITSs (other VRU or non-VRU ITS-S such as vehicles). More explanation regarding TII is provided in clause 6.5.10.9.

The likelihood of a collision associated to the TTC may also be used as a triggering condition for the broadcast of messages (e.g. an infrastructure element getting a complete perception of the situation may broadcast DENM, IVI (contextual speed limit), CPM or MCM).

### 6.5.10.6 Collision risk avoidance

The collision risk avoidance includes the collision avoidance strategy to be selected according to the TTC value.

In the case of an autonomous vehicle, it may consist in the identification of manoeuvre coordination/vehicle motion control to achieve the collision avoidance as per the likelihood of VRU trajectory interception with other road users captured by TII and MI discussed in clause 6.5.10.9.

The collision avoidance strategy also needs to consider environmental conditions:

- Visibility conditions related to the local weather.
- Vehicle stability conditions related to the road state (e.g. slippery).
- Vehicle braking capabilities.

The vehicle collision avoidance strategy then needs to consider the action capabilities of the VRU according to its profile, the remaining TTC, the road and weather conditions as well as the vehicle autonomous action capabilities.

For example, when in good conditions, it is possible to trigger a collision avoidance action when the TTC is greater than two seconds (one second for the driver reaction time and one second to achieve the collision avoidance action). Below two seconds, the vehicle can be considered to be in a "pre-crash" situation and so it needs to trigger an impact mitigation action to reduce the severity of the collision impact for the VRU.

A road infrastructure element such as an RSE may also include a collision risk analysis function as well as a collision risk avoidance function. It may indicate collision avoidance actions to the neighbouring VRUs and non-VRU vehicles. The possible collision avoidance actions (e.g. using MCM as done in the French PAC V2X project) and impact mitigation actions are listed in requirement FSYS08 in clause 5.

The road infrastructure may offer services to support the road crossing by VRU such as traffic lights. When a VRU starts crossing a road at a traffic light level authorizing him, the traffic light should not change of phase as long as the VRU has not completed its crossing. Accordingly, the VAM should contain data elements enabling the traffic light to determine the end of the road crossing by the VRU, e.g. its position.

### 6.5.10.7 Event detection

The event detection function assists the VRU basic service during its operation when transiting from one state to another.

The main events to be considered are:

- Change of a VRU role when a road user becomes vulnerable (activation) or when a road user is not any more vulnerable (de-activation).
- Change of a VRU profile when a VRU enters a cluster with other VRU(s) or with a new mechanical element (bicycle, scooter, moto, etc.), or when a combined VRU breaks up.

At application level, the main events to be considered are:

- Risk of collision between one or several VRU(s) and at least one other VRU (using a VRU vehicle) or a vehicle. Such event is detected via the perception capabilities of the VRU system.

- Change of the VRU motion dynamic (trajectory or velocity) which will impact the TTC and the reliability of the previous prediction.
- Change of the status of a road infrastructure piece of equipment (e.g. a traffic light phase) impacting the VRU movements.

#### 6.5.10.8 Infrastructure services

Existing infrastructure services can be used in the context of the VRU system:

- The broadcast of the SPaT (Signal Phase and Timing) & MAP (SPaT relevance delimited area) is already standardized and used by vehicles at intersection level. In principle they protect VRUs crossing. However, signal violation warnings may exist and can be detected and signalled using DENM. This signal violation indication using DENMs is very relevant to VRU devices as indicating an increase of the collision risk with the vehicle which violates the signal. If its RSE is capable to use local sensors or receive and process VAMs, the traffic light controller may delay the red phase change to green and allow the VRU to safely terminate its road crossing.
- The contextual speed limit using IVI (In Vehicle Information) can be adapted when a large cluster of VRUs is detected (for example, limiting the vehicles' speed to 30 km/hour). At reduced speed, a vehicle may act efficiently when detecting the VRUs by means of its own local sensor system.

#### 6.5.10.9 Manoeuvre coordination

This function executes the collision avoidance actions which are associated to the collision avoidance strategy that has been decided. This function should be present at the vehicle level, depending also on the vehicle level of automation (i.e. not present in non-automated vehicles), and may be present at the VRU device level according to the VRU profile.

At the vehicle level, this function interfaces the vehicle electronics controlling the vehicle dynamic state in terms of heading and velocity.

At the VRU device level, this function may interface the HMI support function, according to the VRU profile, to be able to issue a warning or alert to the VRU according to the TTC.

Manoeuvre coordination can be proposed to vehicles from an infrastructure element, which may be able to obtain a better perception of the motion dynamic of the involved moving objects, by means of its own sensors or by the fusion of their data with the remote perception obtained from standard messages such as CAMs.

The manoeuvre coordination at VRU may be enabled by sharing among the ego-VRU and the neighbouring ITS-S, first the Trajectory Interception Indication (TII) reflecting how likely is the ego-VRU ITS-S trajectory going to be intercepted by the neighbouring ITS-S (other VRU or non-VRU ITSs such as vehicles), and second, Manoeuvre Identifier (MI) to indicate the type of VRU manoeuvre action needed. Both TII and MI are defined in clause 3.1.

Depending upon the analysis of the scene in terms of the sensory as well as shared inputs, simple TII ranges can be defined to indicate the likelihood of the ego-VRU's path to be intercepted by another entity. Such indication helps to trigger timely manoeuvring. For instance, TII could be defined in terms of TII index that may simply indicate the chances of potential trajectory interception (low, medium, high or very high) for collision risk analysis. If there are multiple other entities, the TII may be indicated for the specific entity differentiable via a simple ID which depends upon the simultaneous number of entities in the vicinity at that time. The vicinity could even be just one cluster that the current VRU is located in. For example, the maximum number of entities or users in a cluster is maxClusterSize (see ETSI TS 103 300-3 [i.11], Table 14) per cluster (worst case). However, the set of users that may have the potential to collide with the VRU could be much smaller thus possible to indicate via few bits in say, VAM.

On the other hand, the MI parameter can be helpful in collision risk avoidance by triggering/suggesting the type of manoeuvre action needed at the VRUs. The number of such possible manoeuvre actions may be only a few. For simplicity, it could also define as the possible actions to choose from as {longitudinal trajectory change manoeuvring, lateral trajectory change manoeuvring, heading change manoeuvring or emergency braking/deceleration} in order to avoid potential collision indicated by the TII.

NOTE: The TII and MI parameters can also be exchanged via inclusion in part of VAM message data field structure as outlined in clause 6.9.

## 6.5.11 ITS Station Facilities Layer

### 6.5.11.1 VRU basic service

The VRU basic service is supported by the set of functions illustrated in Figure 8:

- VRU basic service management.
- VRU Cluster management.
- VAM Reception management.
- VAM Transmission management.
- VAM Encoding.
- VAM Decoding.

These functions are described with more details in ETSI TS 103 300-3 [i.11].

The VRU basic service interacts with other Facilities layer functions as represented in Figure 5 through the OFa interface. It also interacts with other entities in the ITS-S through the interfaces defined in ETSI EN 302 665 [1]. These interfaces are further described in ETSI TS 103 300-3 [i.11].

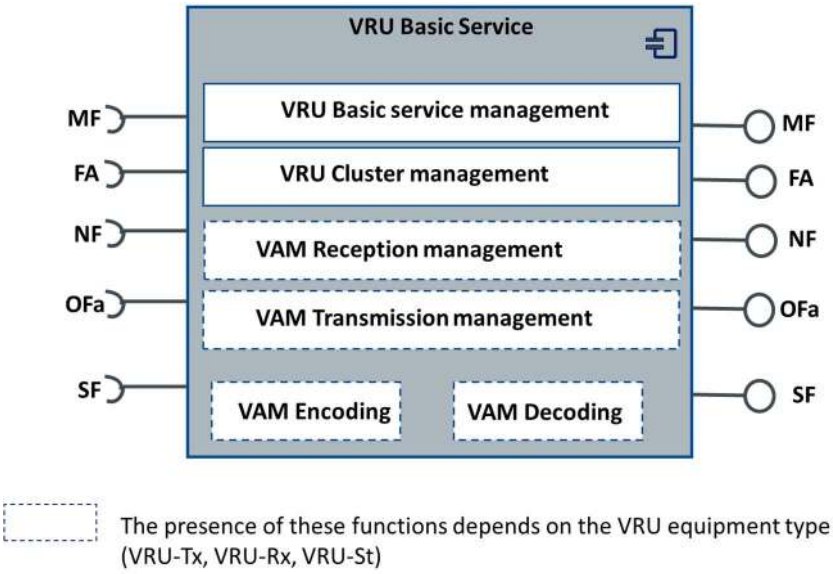



**Figure 8: VRU basic service functional model**

### 6.5.11.2 Local Dynamic Map (LDM)

The Local Dynamic Map (LDM) is the repository of all the dynamic data elements related to the functions supporting the VRU basic service.

The LDM content should be accessible directly from dedicated VRU and vehicle facilities layer functions via an exposed interface (FA SAP: Facilities - Application service access point) from application layer functions. This LDM access is controlled by the LDM management function (see Figure 9). These functions are also able to store and update data elements in the LDM.

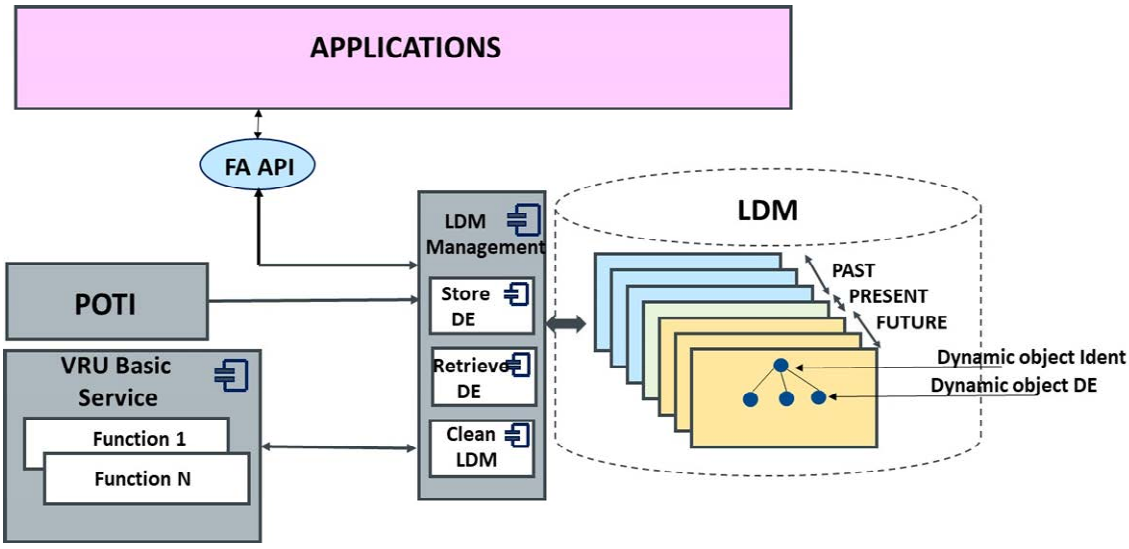



**Figure 9: Example of LDM conceptual model**

Dynamic objects are VRUs, vehicles, road infrastructure objects (e.g. traffic lights), etc. They are identified by a unique identifier and are characterized by dynamic data elements (DE) which need to be updated in real time (e.g. with a sampling period of 100 ms).

Dynamic object characteristics may be kept in the LDM for sufficient time to build the path history of the dynamic object evolutions (vehicles, VRUs) to be able to predict their evolutions in the future (motion dynamic prediction). The motion dynamic prediction can also be performed by navigation function knowing the VRU itinerary and by learning VRU behaviour during frequently used itineraries.

### 6.5.11.3 PoTI

PoTi (Position and Time Management) is specified in ETSI EN 302 890-2 [5].

The PoTi function is present in vehicles' ITS-S and VRU ITS-S. It may also be present in an infrastructure ITS-S and be used to develop a positioning augmentation service.

Position accuracy and confidence level are key performance indicators which are part of PoTi specification.

### 6.5.11.4 Other Application Support Facilities: CA, DEN, CPS, MCS, SPaT, etc.

Other application support facilities functions are responsible to trigger the transmission of their related messages and forward the messages received to the relevant applications.

### 6.5.11.5 Channel Congestion Control (DCC-FAC)

The channel congestion control function acts to optimize the use of the available channel bandwidth during peak traffic, thus avoiding its congestion.

For this purpose, according to current and anticipated channel load, the channel congestion control cooperates with the VRU Basic service function to adjust the VAM transmission to limit the use of the VRU service selected channel to the strict necessity. As a consequence, rules defining transmission triggering conditions may be necessary with the objective to respect the channel congestion control rules.

### 6.5.12 ITS Station Management Entity

#### 6.5.12.1 VRU profile management

The VRU profile management function is an important support element for the VRU basic service as it manages the VRU profile during a VRU active session. The profile management is part of the ITS-S configuration management and is initialized with necessary typical parameters' values to be able to fulfil its operation. The ITS-S configuration management is also responsible for updates (for example: new standard versions) which are necessary during the whole life cycle of the system.

When the VRU basic service role is enabled (vulnerability configured), the VRU profile management needs to characterize a VRU personalized profile based on its experience and on provided initial configuration (generic VRU profile). The VRU profile management may then continue to learn about the VRU habits and behaviours with the objective to increase the level of confidence (reliability) associated with its motion dynamic (trajectories and velocities) and with its evolution predictions.

The VRU profile management is able to adapt the VRU profile according to detected events which can be signalled by the VRU basic service management and the VRU cluster management (cluster creation or cluster break up).

According to its profile, a VRU may or may not be impacted by some road infrastructure event (e.g. evolution of a traffic light phase). Thisenables a better estimation of the confidence level to be associated with his movements. For example, an adult pedestrian will likely wait at a green traffic light for crossing vehicles and then cross the road when the traffic light turns to red. An animal will not take care of the traffic light colour and a child can wait or not according to his age and level of education.

## 6.6 Security

### 6.6.1 Security mechanisms by information flow

Information flows below are identified by the alphanumeric identifiers used in clause 4.5. In the electronic version of the present document, each information flow identifier is a clickable link that links to the security analysis in the relevant part of clause 4.5.

The following information flows/entity activities are implemented with a broadcast transmission pattern and shall be secured with the standard mechanisms defined in ETSI TS 103 097 [6]:

- EA1-1.1, EA2-1.1, EA2-1.2, EA2-2.1, EB1-1.1, EB1-1.2, EB1-1.3, EB1-1.4, EB1-1.5, EB2-1.1, EB2-1.2, EB2-1.3, EB3-1.2, EB3-1.3, EB4-1.1, EB5-1.1, EB5-1.2, EC1-2.1, ED1-1.1, ED1-2.1, ED2-1.1, ED3-1.1, ED3-3.1, ED4-2.1, EF1-1.1, EF1-1.2, EF1-2.1, EF2-2.1, EF2-2.2

The following information flows may be implemented with a broadcast transmission pattern or with a unicast transmission pattern:

- EB3-1.1, EF2-1.1

The following information flows are implemented by a siren and have no communications security mechanism associated with them:

- ED1-2.2, EF1-2.2

The following information flows are implemented as part of communications with a cloud service. For these information flows, the present document does not specify a particular communications security mechanism. However, any communications security mechanism implemented shall be capable of meeting the integrity requirements of the flow. This can be accomplished with standard internet security mechanisms such as Transport Layer Security (TLS) (see IETF RFC 8446 [i.8] and IETF RFC 5246 [i.9]):

- EE1-1.1, EE1-2.1, EE2-1.1, EE2-2.1, EE2-2.2

The following information flows are implemented by sensor perception and communication of that sensor information to an ITS-S. These information flows shall be protected by a persistent communications security mechanism that meets the integrity requirements of the flow. The present document does not specify a particular communications security mechanism to be used:

- EC1-1.1, ED3-2.1, ED4-1.1

## 6.6.2 Roles

### 6.6.2.1 Overall roles

A "role" is an identifier corresponding to a set of activities such that any actor with that role may carry out any of the activities in the set. One goal of the security analysis is to identify natural roles in the system from a security perspective, i.e. collections of activities (as an example: if an actor can do one of the activities maliciously, there is no significant increase in system risk if that actor can do all of those activities maliciously). Other methods to determine roles are possible - for example, a natural way to choose roles is to select those roles to have a correspondence to "real-world" roles within the application - but it is recommended to derive the "natural security roles" as described in this clause so that whatever application roles are eventually selected, the security properties of those roles are properly understood.

The security analysis indicates that the following roles are appropriate:

- VRU:
  - Pedestrian or animal.
  - Person with reduced/impaired mobility.
- Vehicle:
  - Vehicle associated with a rider.
  - Sensor-equipped vehicle (see UC-C1).
- Priority Vehicle.
- Infrastructure device:
  - Infrastructure device equipped with sensors (this will be authorized using the CPS ITS-AID).
- Cloud service.

Accordingly, the final design shall be able to identify the role of each actor in one of the following ways:

- by different ITS-AIDs;
- by different Service Specific Permissions (SSPs) associated with a shared ITS-AID;
- by some other mechanism as appropriate.

### 6.6.2.2 Entitlement to roles

The identification of roles creates two design requirements:

- The design shall provide a mechanism for participants in use cases to determine that counterparties are entitled to act in the appropriate role.
- The design shall provide a mechanism for credential issuance systems (Authorization Authorities (AAs) that issue Authorization Tickets (ATs)) to determine that end-entities requesting ATs are entitled to act in the role requested.

NOTE: AAs and ATs are described in ETSI TS 102 940 [i.10].

### 6.6.3 Pseudonymity

As noted in the security analysis, actors acting in the VRU and Vehicle role require pseudonymity (see also ETSI TS 103 300-3 [i.11]). For actors in both of these cases, for all broadcast information flows, pseudonymity is provided by having multiple certificates.

For actors in the Vehicle role, the messages that are sent and the associated communications patterns are specified in ETSI EN 302 637-2 [3] and ETSI EN 302 637-3 [4]. Pseudonymity (number of certificates, certificate change algorithms and mechanisms) is specified in the same standards or in their reference documents.

For actors in the VRU role, the requirement for pseudonymity should be interpreted taking the following into account:

- VRU devices are even more tightly coupled to individuals than are vehicle-resident devices.
- VRU devices can give away more information about movements than vehicle-resident devices: vehicle-resident devices can show where to park at the mall, but VRU devices can show which exact shop to go into.
- VRUs transmit only intermittently in many scenarios.
- Most VRU devices will have good connectivity to a network.
- Most VRU devices will have the ability to demonstrate to an AA that they are in a known valid state (this is known as "attestation").
- A VRU device will accompany its VRU for a large number of hours per day and may be active or potentially active for a long period of time (during a shopping expedition in a city centre, or while the device owner is out for a long run or bicycle ride).

Based on these properties, the following goals are identified:

- A VRU device should have more flexibility to change certificates than vehicles do under the current certificate policy. For example, a VRU device should be able to change certificates whenever it has been silent for more than 5 s.
- A VRU should be able to request new certificates during the course of its operations, subject to policy about:
  a) whether the VRU device has reached an upper limit on the number of certificates requested;
  b) whether the VRU device has demonstrated that previous certificates have been disposed of (using, for example, the attestation mechanism referred to in the previous list);
  c) further considerations.

### 6.6.4 Misbehaviour Detection

A VRU device that causes false alerts, or a VRU device that creates messages that are physically inconsistent with each other, should be considered for reporting and blacklisting.

## 6.7 Impact of the deployment and automation level roadmap

### 6.7.1 Introduction

The C-ITS platform final report [i.5] has defined a set of C-ITS services for Day 1 and Day 1.5 deployments. The protection of VRUs (pedestrian and cyclists) is considered in the list of Day 1.5 services. In parallel, OEMs are enhancing the vehicle security by increased ADAS capabilities, with a higher level of automation of the vehicles (see SAE J3016 [i.19]) and geo-positioning is getting a higher precision.

In parallel, the horizontal marking and vertical signs are not considered as sufficiently reliable for an automated vehicle to move in a safe automated driving. A redundant method is thus added. It consists in an accurate digital map (twin of the road marking and vertical signs) which should be extended with relevant information dedicated to automated vehicles. The digital map accuracy needs to be better than 10 cm. Accordingly, the vehicle positioning system needs to have a similar level of accuracy to be able to perform an accurate map matching of the mobile object.

NOTE: Map matching is the method used to position a mobile object on the digital map of the vehicle. The horizontal marking and vertical signs are used by automated vehicles to move on the roads. However, these signs are not sufficiently reliable because they are not always visible. Therefore, vehicle manufacturers and their suppliers develop a redundant accurate digital map. But they need a vehicle positioning system which enables to position the vehicle at the lane level (map matching of the vehicle positions). The vehicle also needs to remain in the selected lane especially when turning at an intersection.

The objective of clause 6.7 is to clarify how the VRU functional architecture can be transposed in this roadmap and provide the protection of VRUs starting from the very early deployments.

### 6.7.2 Functional architecture of VRU system with Day 1 services

**Services for Day 1 deployment - assumptions**

These services target vehicle safetyin motorway and urban use cases. They are based on awareness messages (CAM, MAP, SPaT), notifications (DENM) and in-vehicle information.

The C-ITS communication is essentially V2V or I2V.

The level of vehicle automation is expected to be at a low SAE level between 0 and 2 (see SAE J3016 [i.19]). The vehicle is equipped with local sensors only, directly connected to the vehicle on-board system. RSE containing an ITS-S may be equipped with cameras or presence sensors to increase green times at crossings or send contextual speed limit messages (CSM).

**Components of the VRU functional architecture**

Even though VRU protection is not part of Day 1 services, the following components of the VRU system architecture described in Figure 4 are already available in this phase:

- The VRU level is not C-ITS enabled.
- The vehicle level includes the V-ITS-S and the following functions: local perception, motion prediction, HMI and vehicle connected system.
- The roadside level includes the R-ITS-S and the following functions: local perception, mobile objects trajectory prediction, HMI (optional) and RSE connected system.
- The central level includes the C-ITS-S and all the functions displayed at this level in Figure 4.

**Methods to detect VRU at risk**

The risk of a potential collision shall be detected using one of the following methods:

- In vehicle, detection by a local sensor interacting with the V-ITS-S.
- In vehicle, reception of a CAM [3] or DENM [4] signalling the presence of VRUs (pedestrians, cyclists) on the road.
- In infrastructure (roadside and central level), detection of VRUs in designated areas (zebra crossing, road), while vehicles are authorized on the same areas (and vice versa).

NOTE: Detecting a VRU at risk requires to be able to predict reliably the motion dynamic of the VRU as soon as possible. In most cases, this is not possible before a short TTC because of the large number of possible trajectories and velocities at the VRU level. To increase the TTC and avoid either false positive information or very late valid positive information, it is then possible to rely on other functions such as the VRU navigation system, or machine learning system memorizing VRU habits and behaviour from repetitive motion dynamic.

**Methods to signal VRU at risk or prevent the risk**

The risk of a potential collision with a VRU shall be notified or mitigated using one of the following methods:

- Direct warning to the vehicle driver through the HMI.

- DENM sent by a R-ITS-S or by a V-ITS-S to the neighbouring ITS-S.
- Action from an R-ITS-S on a traffic light controller.

### 6.7.3 Functional architecture of VRU system with Day 1.5 services

**Services for Day 1.5 deployment - assumptions**

The roadmap of service deployment is incremental. Accordingly, the Day 1 services are present in this phase. Additional services of interest are the CPS and the VRU basic service. New messages are available to the ITS stations: CPM and VAM.

The C-ITS communications additionally include VRU2VRU, V2VRU/VRU2V as well as I2VRU/VRU2I.

The level of vehicle automation is at a medium SAE level between 2 and 4 (see SAE J3016 [i.19]). Actions can be performed on the vehicle, C-ITS warnings can be displayed on the on-board system, but the driver has still the control of the vehicle. RSE containing an ITS-S may be equipped with cameras or presence sensors and actively participate to the C-ITS traffic safety.

**Components of the VRU functional architecture**

The following components of the VRU system architecture described in Figure 4 will be available in this phase:

- The VRU level includes all the functions present in Figure 4.
- The vehicle level includes all the functions present in Figure 4, except for the vehicle motion control function.
- The roadside level includes all the functions present in Figure 4.
- The central level includes all the functions present in Figure 4.

**Methods to detect VRU at risk**

The risk of a potential collision shall be detected using one of the following methods:

- All methods used with Day 1 services.
- In the VRU device, reception of a CAM or DENM signalling the presence of a vehicle crossing the VRU predicted trajectory.
- In the VRU device, reception of a VAM signalling the presence of a VRU crossing the VRU predicted trajectory. However, this does not apply when the two VRUs belong to Profile 1 (see clause 6.1).
- In vehicle, reception of a CPM signalling the presence of a vehicle crossing the VRU predicted trajectory. This may be correlated with information received from local sensors.
- In infrastructure (roadside and central level), analysis of predicted trajectories of the mobile objects and identification of a potential collision.

**Methods to signal VRU at risk or prevent the risk**

The risk of a potential collision with a VRU shall be notified or mitigated using one of the following methods:

- All methods used with Day 1 services.
- Direct warning to the VRU through the HMI or through an HMI located in the R-ITS (for example a beep from the traffic light).
- VAM sent by the ITS-S in the VRU device.
- CPM sent by a R-ITS-S or a V-ITS-S to the neighbouring ITS-S.
- Action on the vehicle depending on its level of automation (slowing down, braking, changing lane, etc.).

### 6.7.4 Functional architecture of VRU system with beyond-Day 1.5 services

**Services for beyond-Day 1.5 deployment - assumptions**

The roadmap of service deployment is incremental. Accordingly, the Day 1.5 services are present in this phase. As this is not yet fully defined, the expectation is that all planned C-ITS services are available, for example the MCM.

The C-ITS communications additionally includes all types of exchanges as defined in Figure 1.

The level of vehicle automation is from a medium to high (i.e. fully automated) SAE level between 3 and 5 (see SAE J3016 [i.19]). Automated actions can be performed on the vehicles, warnings can be displayed on the on-board system, but the driver may not have the control of the vehicle. RSEs containing an ITS-S actively participate in the C-ITS traffic safety.

**Components of the VRU functional architecture**

All the components of the VRU system architecture described in Figure 4 will be available in this phase.

**Methods to detect VRU at risk**

The risk of a potential collision shall be detected using one of the following methods:

- All methods used with Day 1.5 services.

**Methods to signal VRU at risk or prevent the risk**

The risk of a potential collision with a VRU shall be notified or mitigated using one of the following methods:

- All methods used with Day 1.5 services.
- MCM sent by a R-ITS-S or a V-ITS-S to the neighbouring ITS-S.
- If the level of automation of the vehicle is high and the driver does not have the control of the vehicle, it may be questionable to display a direct warning to the vehicle driver through the HMI, but this is out of scope of the present document.

## 6.8 Interfaces between entities

### 6.8.1 Introduction

The interfaces between different ITS stations of different levels of the architecture, i.e. external interfaces, and those between different functional components of each level, i.e. internal interfaces, are illustrated in Figure 4. The interfaces between the VRU Basic Service and other facilities layer entities in the ITS-S architecture are illustrated in Figure 5.

### 6.8.2 External interfaces

Parameters of the external interfaces shall include the descriptive, physical and dynamic state information of a VRU so that the collision risk can be estimated and warned, and the strategy for collision avoidance can be deduced. The parameters may also include additional information of a VRU cluster for efficient network traffic.

**Table 31: External interfaces**

| Interface | Description |
|---|---|
| VRU ITS Station ←→ Vehicle ITS Station | CAM and VAM messages are sent bidirectional to exchange information on type of road user, absolute position, velocity, etc. between nearby road users. For a collision risk warning at intersections, a Vehicle ITS Station can send a DENM message with intersection collision warning for left/right turning or crossing scenarios. |
| VRU ITS Station ←→ Roadside ITS Station | VAM messages are broadcasted by VRU ITS Stations and can be used by the Roadside ITS Stations. The Roadside ITS Stations can send warnings on signal violations or intersection collision risk via DENM. The Roadside ITS Stations can send CPMs regarding the presence of VRUs. |
| VRU ITS Station ←→ Central ITS Station | This interface can be used for applications to exchange non-geographical related information via a central system. The central ITS-S can detect a risk of collision by using VAM sent by a VRU ITS-S and information from vehicles. |
| Vehicle ITS Station ←→ Roadside ITS Station | CAM messages are broadcasted by Vehicle ITS Stations and can be used by the Roadside ITS Stations to estimate the collision risk. The Roadside ITS Stations can send warnings on signal violations or intersection collision risk via DENM. Assistance messages via SPaT for turning and/or crossing where a VRU ITS Station can be present along the trajectory (as defined in ETSI TS 103 301 [i.13]) are focussing on I2V. Roadside ITS-S may also broadcast CPMs (awareness) or MCMs (automated vehicle motion control) regarding the presence of VRUs. |
| Vehicle ITS Station ←→ Central ITS Station | This interface is used to exchange appropriate ETSI C-ITS messages specified for the Vehicle ITS Station and Central ITS Station. |
| Roadside ITS Station ←→ Central ITS Station | This interface is used to exchange appropriate ETSI C-ITS messages specified for the Roadside ITS Station and Central ITS Station. |
| Vehicle ITS Station ←→ Vehicle ITS Station | This interface is used to disseminate CPMs regarding the presence of VRUs. A vehicle perceiving a VRU may broadcast a CPM signalling the VRU to other vehicles. If a vehicle receiving a CPM identifies a risk of collision with the signalled VRU, a collision avoidance action can be triggered. |

Further details for the VAM message are provided in clause 6.9 and in ETSI TS 103 300-3 [i.11].

### 6.8.3 Internal interfaces

Data exchanged via the internal interfaces of the VRU system are used to generate the parameters to be sent via the external interfaces. They are also used to estimate/predict the collision risk and avoid the predicted collision based on the parameters received via the external interfaces.

Table 32 describes the internal interfaces at VRU level.

**Table 32: Internal interfaces at VRU level**

| From | To | Parameters |
|---|---|---|
| VRU device sensor system | Sensor data fusion and actuator<br>Local perception<br>VRU ITS Station | Data obtained from various local sensors of the VRU level. |
| Sensor data fusion and actuator | Motion prediction<br>Local perception<br>VRU ITS Station | Fusion data from information obtained from various local sensors of the VRU level. |
| Local Perception | VRU ITS Station | Perception data processed from information obtained from various local sensors of the VRU level. |
| Motion prediction | VRU ITS Station | Manoeuvre prediction of the VRU level e.g. reflected by the two parameters TII and MI as defined in clause 3.1 and as discussed in clause 6.5.10.9. |
| VRU ITS Station | VRU connected system and HMI | Information of the VRU ITS Station to advise or instruct the associated road user such as a pedestrian, cyclist or PTW driver via an external HMI (e.g. smartphone). |

Table 33 describes the internal interfaces at vehicle level.

**Table 33: Internal interfaces at vehicle level**

| From | To | Parameters |
|---|---|---|
| Vehicle sensor system | Sensor data fusion and actuator<br>Local perception<br>Vehicle ITS Station | Data obtained from various local sensors of the vehicle level. |
| Sensor data fusion and actuator | Motion prediction<br>Local perception<br>Vehicle ITS Station | Fusion data from information obtained from various local sensors of the vehicle level. |
| Local perception | Vehicle ITS Station | Perception data processed from information obtained from various local sensors of the vehicle level. |
| Motion prediction | Vehicle ITS Station<br>Vehicle motion control | Manoeuvre prediction of the vehicle level. |
| Vehicle ITS Station | Vehicle motion control | Information to control the vehicle level. |
| Vehicle ITS Station | Vehicle connected system and HMI | Information of the Vehicle ITS Station to advise or instruct the associated road user such as a vehicle driver and motorcycle driver via an external HMI. |

Table 34 describes the internal interfaces at roadside level.

**Table 34: Internal interfaces at roadside level**

| From | To | Parameters |
|---|---|---|
| Roadside sensor system | Sensor data fusion and actuator<br>Local Perception<br>Roadside ITS Station | Data obtained from various local sensors of the roadside level. |
| Sensor data fusion and actuator | Mobile object prediction<br>Local Perception<br>Roadside ITS Station | Fusion data from information obtained from various local sensors of the roadside level. |
| Local perception | Roadside ITS Station | Perception data processed from information obtained from various local sensors of the roadside level. |
| Mobile object prediction | Roadside ITS Station | Manoeuvre prediction of mobile objects. |
| Roadside ITS Station | RSE connected system and HMI | Information of the VRU ITS Station to advise or instruct the associated road user such as a roadside operator. |

Table 35 describes the internal interfaces at Central level.

**Table 35: Internal interfaces at Central level**

| From | To | Parameters |
|---|---|---|
| Sensor | TMS (Traffic Management System) | Data obtained from various local sensors of the central level. |
| TMS (Traffic Management System) | Central Traffic Management Console and HMI Local | Traffic and road state information, i.e. the actual status of flow/velocity/travel times and measures, warnings and status of traffic signs. |
| Central Traffic Management Console and HMI | Central ITS Station | Information on (road works) warnings or route advice to cooperative vehicles via Central ITS Station - optionally Roadside ITS Station - Vehicle ITS Station. Optionally dynamic road signalling information could be sent over this interface. |
| Central ITS Station<br>Central Traffic Management Console and HMI | Internet Information System | Specific information or request for specific information for internet-based applications. |
| Internet Information System | Central ITS Station<br>Central Traffic Management Console and HMI | Specific information or response for specific information for internet-based applications. |

Further details for the parameters of the internal interfaces are described in clauses 6.5.2 to 6.5.8.

### 6.8.4 VRU Basic Service interfaces

The interactions between the VRU Basic Service and other facilities layer entities in the ITS-S architecture are used to obtain information for the generation of the VAM. The interfaces for these interactions are described in Table 36 and in ETSI TS 103 300-3 [i.11].

**Table 36: VRU Basic Service interfaces**

| Interfaced functionality | Parameters |
|---|---|
| PoTi | Information of the positioning and timing are sent to the VRU basic service. Further details are described in clause 6.5.10.3. |
| Congestion control | Information to optimize the use of the available channel are sent to the VRU basic service. Further details are described in clause 6.5.10.5. |
| HMI support | Information to be exchanged with a VRU. Further details are described in clause 6.5.7. |
| LDM | LDM/VAM data are exchanged between LDM and the VRU basic service.. Further details are described in clause 6.5.10.2. |
| Other application support facilities | Information to trigger the transmission of messages are sent to the VRU basic service. The VRU basic service forwards received messages to the relevant applications. Further details are described in clause 6.5.10.4. |
| Device Data Provider (DDP) | The DDP provides the device status information obtained from its local perception entities to the VRU basic service. |
| CA Basic service | In case of a motorcycle, the VRU basic service needs to inform the CA basic service that the vehicle is a VRU from VRU profile 3 and trigger the dedicated container when transmitting CAMs. |

## 6.9 Vulnerable Road User Awareness Message, VAM

The message specified for the VRUs awareness (VAM) shall be harmonized in the largest extent with the existing Cooperative Awareness Messages (CAM) defined in ETSI EN 302 637-2 [3]. The transmission of the VAM shall be limited to the VRU profiles specified in clause 6.1.

The Vulnerable Road User Awareness Message (VAM) shall contain all required data depending on the VRU profile and the actual environmental conditions. The data elements in the VAM should be at least as described in Table 37.

NOTE: The list below is a preliminary list of DEs based on the specifications in the present document. The full specification of the VAM can be found in clause 7 of ETSI TS 103 300-3 [i.11].

**Table 37: VAM data elements**

| Parameter | Insertion in VAM | Comments |
|---|---|---|
| VAM header including VRU identifier | M | |
| VRU position | M | |
| Generation time | M | |
| VRU profile | C | VRU profile and sub-profile are included with a lower period than dynamic DEs of the VAM<br>See ETSI TS 103 300-3 [i.11] |
| VRU sub-profile | C | e.g. VRU profile is pedestrian, VRU sub-profile is infant, adult, child, road worker, etc. |
| VRU cluster identifier | O | |
| VRU cluster position | O | |
| VRU cluster dimension | O | geographical shape and size |
| VRU cluster cardinality size | O | number of members in the cluster |
| VRU size class | C | mandatory if profile and sub-profile are included in the VAM |
| VRU speed | M | |
| VRU direction | M | |
| VRU orientation | O | |
| Predicted trajectory | O | succession of way points |
| Heading change indicators | O | turning left or turning right indicators |

| Parameter | Insertion in VAM | Comments |
|---|---|---|
| Acceleration change indicator | O | |
| NOTE: "M" stands for "mandatory" which means that the data element shall be always included in the VAM message. "O" stands for "optional" which means that the data element can be included in the VAM message. "C" stands for "conditional" which means that the data element shall be included in the VAM message under certain conditions. | | |

# 7 Impact on existing standards and protocols

## 7.1 Introduction

The recommendations to update the existing C-ITS standards and protocols for the messages used by the VRU applications are presented below.

Furthermore, a specification is needed for the VRU Basic Service, specifying:

- VAM generation rules and triggering management.
- VAM RX and VAM TX.
- VAM encoding and construction.
- VAM decoding and decomposition.

## 7.2 Facilities Layer

- In the Facilities Layer [i.12], [i.13], [i.14] and [2]:
  - Introduce VRU basic service in the Facilities layer structure, functional requirements and specifications ETSI TS 102 894-1 [2].
  - Add new definitions of data elements in the Common Data Dictionary (CDD, ETSI TS 102 894-2 [i.14]) if necessary.
  - PoTI:
    - Add specification of VRU clusters reference point.
  - LDM:
    - Consider VAM.
  - Other Application Support Facilities: CA, DEN, CPS, MCS, SPaT, etc.:
    - CA: Add a s special vehicle container to signal VRUs from VRU Profile 3 which do not send the VAM.
    - DEN: Detail the cause code values according to the VRU profile (for example, the sub-cause codes of humanPresenceOnTheRoad (12), collisionRisk (97)). Triggering conditions should be revised as well.
    - CPS: Align the classification of VRUs according to the profiles specified in clause 6.1.
    - MCS: Future use of VAM information in the MCS processing for reliability enhancements.
    - SPaT: processing of SPaT information received by the VRU for awareness and e.g. red-light violation warning.
    - MAP: High resolution map of actual crossing (better than 10 cm), identify sidewalks, bicycle lanes and pedestrian areas in the maps.

## 7.3 Networking & Transport Layer

The VAM and the related functionalities are Networking & Transport layer agnostic. No specific recommendation and impacts on the Networking & Transport layer are expected, beside the allocation of a specific BTP port number for the VRU basic service in ETSI TS 103 248 [i.27].

For GeoNetworking a higher resolution of the addressing might be needed. In addition, a separate cluster addressing might be needed.

## 7.4 Access Layer

The VAM and the related functionalities are access layer agnostic. No specific recommendation and impacts on the access layer are expected.

NOTE: No specific VRU related functionalities are foreseen. Nevertheless, a better positioning capability of the access layer as provided by IEEE 802.11bd [i.25] might be useful.

## 7.5 Management Entity

Functions in the Management entity should also include the VRU profile management.

## 7.6 Security Entity

No changes seem to be necessary to the security entity as specified in ETSI EN 302 665 [1].

# 8 Conclusion

The present document has introduced a complete VRU ITS system concept including:

- safety analyses of the VRU use cases in clause 4.5;
- requirements set for the use with VRUs, combined VRUs and VRU clusters in clause 5;
- a functional architecture specification in clause 6; and
- the relevant interfaces and interactions with other relevant ITS standards and entities in clause 7.

The system concept is based on the use cases defined in ETSI TR 103 300-1 [i.1].

The originally proposed VRU profiles in ETSI TR 103 300-1 [i.1] are specified in clause 6.1 of the present document.

# Annex A (informative): Change History

| Date | Version | Information about changes |
|---|---|---|
| April 2021 | 2.2.1 | Clarification and improvement of functional requirements, cluster concept, and functional architecture of the VRU system. |

# History

| Document history | | |
|---|---|---|
| V2.1.1 | May 2020 | Publication |
| V2.2.1 | April 2021 | Publication |
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